

2 Demand Theory

This chapter expands on two elements analyzed in chapter 1—utility functions and feasibility constraints—as the ingredients of the consumer’s utility maximization problem (UMP). Recall that the consumer chooses a bundle that maximizes his utility given the economic and physical constraints he faces. The bundle that emerges from such maximization problem constitutes the demand function of the individual (in particular, the Walrasian demand), which we study in section 2.2. The Walrasian demand describes how optimal purchases of a certain good by an individual are affected by variations in the price of this good, changes in the prices of other goods, and changes in his income. In addition we can evaluate the individual’s utility function at the Walrasian demand (i.e., at the bundle that maximizes the individual’s utility) in order to identify the height that his utility reaches at the optimum. This utility evaluation is often referred to as the individual’s “indirect utility function,” which is the content of section 2.3. In sections 2.4 and 2.5 we switch our attention to the consistency requirement on individual choices that we explored in chapter 1, the WARP, but present it in terms of the Walrasian demand functions, along with the implications that WARP has in terms of comparative statics, namely when the price of a good changes. Section 2.6 further analyzes the comparative statics of varying the price of one good, and how such effect can be divided into two separate effects: the income and substitution effects, as captured by the Slutsky equation.

Sections 2.7 and 2.8 examine the dual of the UMP, the expenditure minimization problem (EMP), in which the individual seeks to minimize his total expenditure on goods and services conditional on reaching a specific utility level. The bundle resulting from the EMP is the so-called Hicksian demand, whereas the minimal expenditure required to purchase such a bundle is commonly referred to as the expenditure function. Last, in the remaining sections of the chapter we explore the relationships between the different results of the UMP and the EMP, such as the connection between the Walrasian and the Hicksian demands, and the relationship between the

indirect utility function and the expenditure function. These connections can be useful in applications where we have information about elements of, for instance, the EMP, but seek to infer elements from the UMP. The appendices to this chapter elaborate on these connections by analyzing under which conditions the UMP is the dual of the EMP.

2.1 The Utility Maximization Problem

We are now ready to combine individual preferences, embodied in a consumer's utility function, and his budget set, which represents the set of bundles he can afford. In particular, the consumer maximizes his utility level by selecting a bundle x (the choice variable of his maximization problem, where x can be a vector) subject to his budget constraint $p \cdot x \leq w$. Formally, his UMP can be represented as follows:

$$\max_{x \geq 0} u(x) \quad \text{subject to } p \cdot x \leq w \quad (\text{UMP}).$$

For compactness, we denote the solution/s to the UMP as the argmax of UMP,¹ and represent it as $x(p, w)$: the Walrasian demand that, for a given price vector and wealth level, specifies a bundle of goods that maximizes the consumer's utility.²

Figure 2.1 depicts the Walrasian demand $x(p, w)$ at bundle A , where the consumer reaches a utility level of u_2 by exhausting all his wealth. Notice that bundle B and C are not optimal because, despite exhausting the consumer's wealth, they yield a lower utility level u_1 , where $u_1 < u_2$. Bundle D is unaffordable, and hence it cannot be the argmax of the UMP given a wealth level of w .

2.1.1 Existence

One important point is whether the maximization problem above has a solution. The Weierstrass theorem provides us with a positive answer (see the mathematical appendix for more details on this theorem): if the objective function we maximize (utility function) is continuous and the budget constraint defines a closed and bounded set (given that $p \gg 0$ and $w > 0$), the problem has a solution. In addition, the solution is unique if preferences are strictly convex. We return to this issue below. Interestingly, if the utility function is continuous and preferences satisfy LNS over the consumption set $X = \mathbb{R}_+^N$, then the Walrasian demand $x(p, w)$ satisfies:

¹ Recall that the argmax refers to the argument, x , that solves a given maximization problem.

² Some textbooks refer to the solution/s of the UMP as the "Marshallian demand," which coincides with the Walrasian demand identified above.

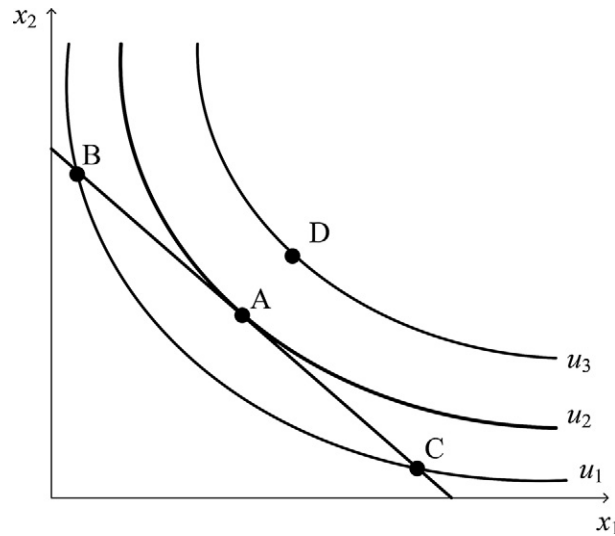


Figure 2.1
Utility maximization problem

1. *Homogeneity of degree zero* $x(p, w) = x(\alpha p, \alpha w)$ for all $p \gg 0$, $w > 0$, and for all $\alpha > 0$.
2. *Walras's law* $p \cdot x(p, w) = w$.
3. *Convexity/uniqueness* If preferences are convex, then the Walrasian demand $x(p, w)$ defines a convex set. If preferences are *strictly* convex, then the Walrasian demand $x(p, w)$ contains a single element.

We next elaborate on each of these three properties.

2.1.2 Homogeneity

First, note that homogeneity of degree zero should come as no surprise. Specifically, an increase in both the price vector and wealth level does not change the consumer's budget set.³ In particular, the budget set satisfies

³ Remember that a function is homogeneous of degree r if increasing all the arguments of the function by a common factor α produces a change in the value of the function of α^r . Hence, when a function is homogeneous of degree zero, an increase in all its arguments by a common factor α produces a change of $\alpha^0 = 1$ in the value of the function, thus not modifying the initial value of the function, $u(\alpha x_1, \alpha x_2) = u(x_1, x_2)$. In the case of individual's Walrasian demand, his optimal consumption bundle before scaling all prices and wealth by a common factor α coincides with his optimal consumption bundle after this scaling takes place.

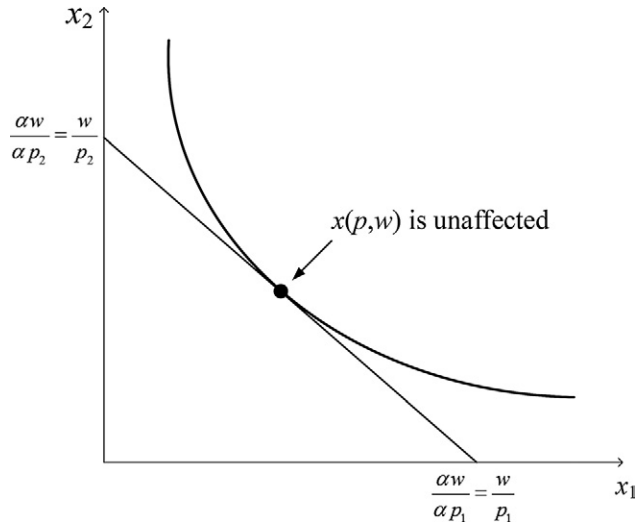


Figure 2.2
 $x(p, w)$ homogeneous of degree zero

$$\{x \in \mathbb{R}_+^L : px \leq w\} = \{x \in \mathbb{R}_+^L : \alpha px \leq \alpha w\}.$$

Figure 2.2 depicts a budget line $B_{p,w}$ for two goods, with intercepts w/p_1 and w/p_2 , and illustrates how a common increase α in price and wealth produces a budget line $B_{\alpha p, \alpha w}$ with exactly the same intercepts $\alpha w / \alpha p_i = w/p_i$ for both goods $i = \{1, 2\}$ as budget line $B_{p,w}$. Hence budget lines $B_{p,w}$ and $B_{\alpha p, \alpha w}$ exactly coincide.⁴ Since the budget set is unchanged, the optimal bundle selected by the individual, $x'(p, w)$, should not change either.

2.1.3 Walras's Law

Walras's law follows from LNS. Indeed, if the consumer were to select a bundle that lies strictly inside the budget set (so that he is not exhausting his wealth, as in bundle x in figure 2.3), we could find another bundle y at ε -distance from bundle x that is strictly preferred by the individual to bundle x . In this case, however, the initial bundle x would not be utility maximizing because there are other bundles that are still

⁴ Note that Walrasian demand $x(p, w)$ satisfies homogeneity of degree zero even when preferences are linear (goods are regarded as substitutes), yielding a unique (or a continuum) of optimal consumption bundles that would be unaffected by a common change in both prices and wealth. A similar argument applies for preferences for complements.

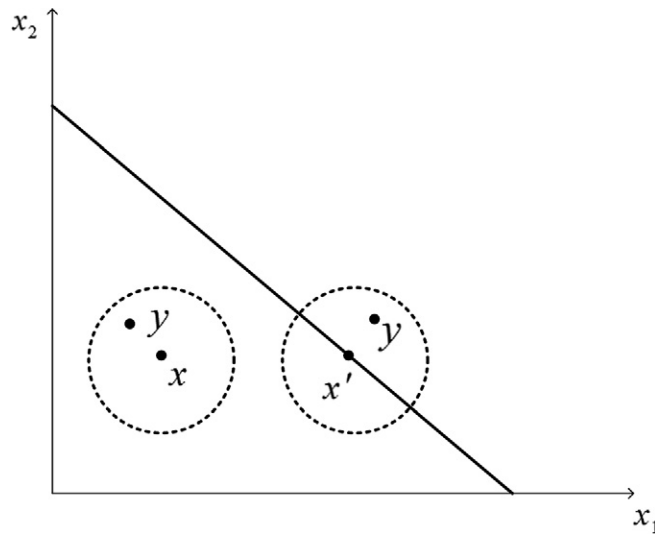


Figure 2.3
Walras's law

affordable and which are strictly preferred by the consumer. If the bundle the consumer chooses lies, in contrast, on the budget line (as bundle x' in figure 2.3), we could still identify bundles that are strictly preferred to x' , such as those to the northeast of the budget line. However, these bundles would be unaffordable to the consumer, and thus cannot be utility maximizing. In summary, utility-maximizing bundles, such as x' , must lie on the budget line, thus exhausting all wealth.

2.1.4 Convexity

If preferences are convex (but not strictly convex), a continuum of bundles can be utility maximizing. In particular, the budget line and the indifference curve can become tangent along several bundles (see left-hand panel of figure 2.4), forming a convex set. Indeed, if we construct the convex combination of any two bundles in this set, the resulting bundle also lies within the set. If, in contrast, the consumer's preferences are strictly convex, he selects a unique bundle as his Walrasian demand (unique tangency point between the budget line and the indifference curve, as depicted in the right-hand panel of figure 2.4).

Having described the UMP, we can now examine the first-order (necessary) conditions of this maximization problem:

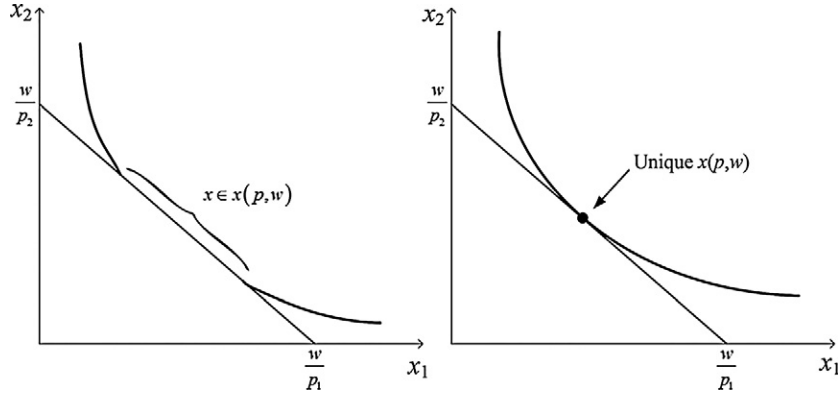


Figure 2.4
Convex and strictly convex preferences

$$\max_{x \geq 0} u(x) \quad \text{subject to } p \cdot x \leq w \quad (\text{UMP}).$$

We can solve the UMP equation given above by setting up the Lagrangian⁵

$$L = u(x) + \lambda [w - p \cdot x].$$

Taking first-order conditions, we obtain

$$\frac{\partial L}{\partial x_k} = \frac{\partial u(x^*)}{\partial x_k} - \lambda p_k \leq 0 \quad \text{for every good } k, \text{ with equality if } x^* > 0, \text{ and}$$

$$\frac{\partial L}{\partial \lambda} = w - p \cdot x^* \leq 0.$$

That is, in the interior optimum, $\partial u(x^*)/\partial x_k = \lambda p_k$ for every good k . Solving for λ for two goods l and k , we obtain

$$\frac{\partial u(x^*)/\partial x_l}{p_l} = \frac{\partial u(x^*)/\partial x_k}{p_k}.$$

Intuitively, this optimality condition can be understood as that the consumer allocates his consumption among goods l and k until the point where the marginal utility per dollar spent on good l , MU_l/p_l , coincides with that of good k , MU_k/p_k . Informally, the consumer distributes his consumption to achieve the same “bang for the buck” across goods.

⁵ See, for instance, chapter 18 in Simon and Blume’s (1994) textbook.

Rearranging, the above optimality condition can be alternatively expressed as

$$\frac{\partial u(x^*)/\partial x_l}{\partial u(x^*)/\partial x_k} = \frac{p_l}{p_k} \Leftrightarrow MRS_{l,k} = \frac{p_l}{p_k}$$

In words, the consumer allocates consumption among goods l and k until the point where his willingness to substitute across goods (his MRS) coincides with the rate at which he can substitute goods in the market (as captured by the price ratio). Graphically, optimality implies that the slope of the consumer's indifference curve coincides with that of the budget line.

2.1.5 Sufficiency

A natural question at this point is whether the necessary conditions above are also sufficient. In other words, under which conditions can we guarantee that the Walrasian demand is the maximum of the UMP (and not the minimum!). In particular, this is the case when (1) the utility function is quasi-concave and monotone, and (2) the vector of first-order derivatives is different from zero for all x , that is, $\nabla u \neq 0$ for all x . Let us briefly analyze these conditions. First, the condition stating that the utility function should be monotone only implies that, if we increase both goods simultaneously, we reach a higher utility level, and that if we increase the amount of some (but not all) goods, the consumer is not made worse off, which is expected in most economic applications. If, instead, the utility function does not satisfy monotonicity, the consumer's indifference map could look like that in figure 2.5. In this setting, the consumer would choose bundle A (at the corner), since it yields the largest utility level, u_3 , given his budget. At point A , however, the tangency condition $MRS_{1,2} = p_1/p_2$ does not hold, since the consumer would like to further decrease his consumption of x_1 .⁶

Second, the condition that the first-order derivatives are different from zero simply guarantees that there are no bliss points. Intuitively, if the vector of first-order derivatives was zero, we would have reached the “peak” of utility, which is the highest point of the “utility mountain” in our 3D representation in chapter 1. At this point, however, the individual would not be able to find any other more preferred bundle, thus violating LNS (and we know that monotone utility functions must satisfy LNS). Finally, the condition that the utility function satisfies quasi-concavity is also easy to justify. Figure 2.6 represents an indifference map of an individual whose preferences violate quasi-concavity.

⁶ At point A , the indifference curve is flatter than the budget line, implying that $(MU_1/MU_2) < (p_1/p_2)$, or $(MU_1/p_1) < (MU_2/p_2)$. Hence the marginal utility per dollar spent on good 2 is larger than that on good 1, inducing the consumer to purchase units of good 2 alone.

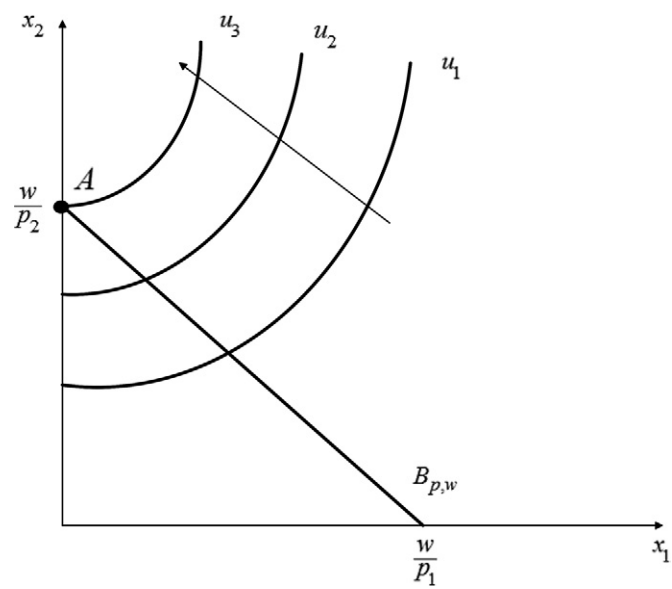


Figure 2.5
Corner solution

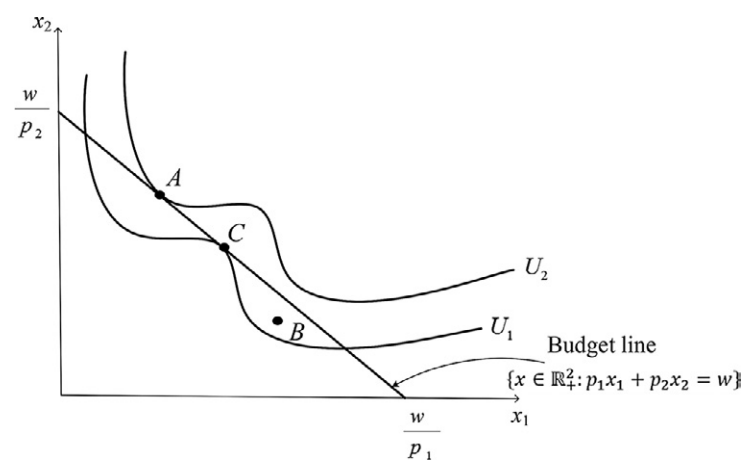


Figure 2.6
UMP with non-quasi-concave utility function

As the figure shows, the *UCS* of bundle *C* is not convex. This implies that the tangency condition between the indifference curves and the budget line is not a sufficient condition for a utility-maximizing bundle. Specifically, a point of tangency such as bundle *C* gives a lower utility level than a point of nontangency such as bundle *B*. Therefore, if preferences do not satisfy quasi-concavity, the Kuhn–Tucker conditions (graphically represented by the tangency condition) are not sufficient for a maximum. The true maximum in this case is actually bundle *A*. Because the three requirements for the necessary conditions to become sufficient are relatively mild (utility function being quasi-concave, monotonic, and $\nabla u \neq 0$), we can then expect Kuhn–Tucker conditions to be sufficient in many economic applications.

Figure 2.7 represents an interior solution where the utility maximizing bundle satisfies $MRS_{l,k}/p_l/p_k$, graphically implying that the slope of the budget line coincides with the slope of the indifference curve evaluated at the optimum.

Example 2.1: UMP with the Cobb–Douglas utility function Let us consider an example of a utility maximization problem. Take the Cobb–Douglas function expressed by $u(x, y) = x^\alpha y^\beta$, which is subject to the following budget constraint $p_x x + p_y y = w$, where, for convenience, we assume that $\alpha + \beta = 1$. We can now solve for the utility maximizing values of x and y for any positive prices (p_x, p_y) and wealth (w). Setting up the Lagrangian, we write

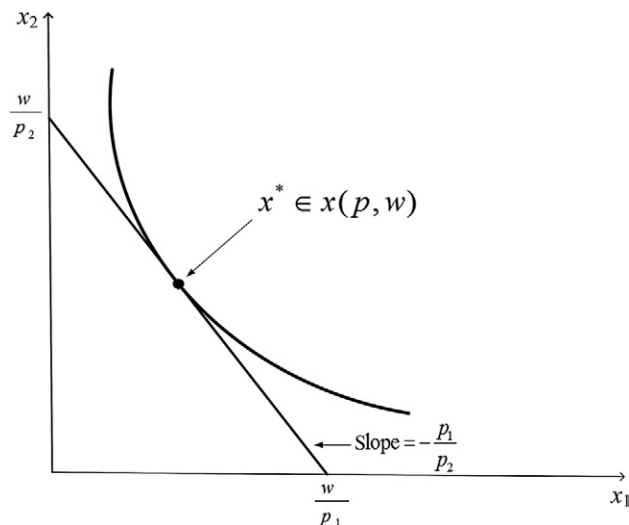


Figure 2.7
Interior solution

$$L(x, y; \lambda) = x^\alpha y^\beta + \lambda(w - p_x x - p_y y),$$

which yields the first-order conditions

$$\frac{\partial L}{\partial x} = \alpha x^{\alpha-1} y^\beta - \lambda p_x = 0,$$

$$\frac{\partial L}{\partial y} = \beta x^\alpha y^{\beta-1} - \lambda p_y = 0,$$

$$\frac{\partial L}{\partial \lambda} = w - p_x x - p_y y = 0.$$

Taking the ratio of the first two terms, we obtain $\alpha y / \beta x = p_x / p_y$, or

$$p_y y = \frac{\beta}{\alpha} p_x x = \frac{1-\alpha}{\alpha} p_x x,$$

where the final equation follows because $\alpha + \beta = 1$, i.e., $\beta = 1 - \alpha$. Substituting the first-order conditions into the budget set $p_x x + p_y y = w$ gives

$$p_x x \left(1 + \frac{1-\alpha}{\alpha} \right) = \frac{1}{\alpha} p_x x = w.$$

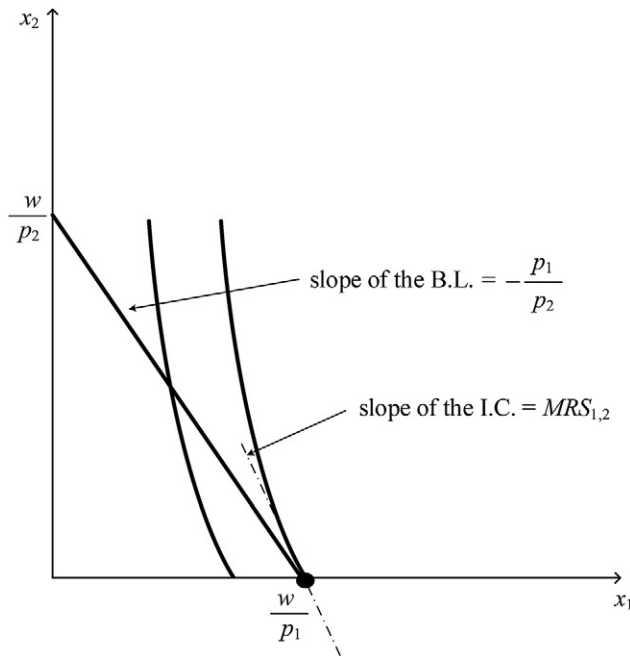
Finally, solving for x yields the Walrasian demand for good x , $x(p_x, p_y, w) = \alpha w / p_x$. A similar set of manipulations gives the Walrasian demand for good y , $y(p_x, p_y, w) = \beta w / p_y$. The demand for every good k is increasing in wealth, w , decreasing in its own price, p_i , but unaffected by the price of the other good, p_l , where $l \neq k$. ■

2.1.6 Corner Solutions

We have so far analyzed interior solutions. If, however, the individual prefers to consume zero amounts of some of the goods, the preceding tangency condition will not be satisfied. In particular, at a corner solution we find that, after taking first-order conditions, $MRS_{l,k} > p_l / p_k$, or alternatively,

$$\frac{\partial u(x^*) / \partial x_l}{p_l} > \frac{\partial u(x^*) / \partial x_k}{p_k}$$

Intuitively, the marginal utility per dollar spent on good l is larger than that on good k . As a consequence, even after spending all his income on good l (at the corner), the consumer would still like to consume more units of good l . In the first-order conditions, this entails that $\partial u(x^*) / \partial x_k \leq \lambda p_k$ for those goods whose consumption is zero, $x_k^* = 0$, that is, $\partial u(x^*) / \partial x_k / p_k \leq \lambda$; and $\partial u(x^*) / \partial x_l = \lambda p_l$ for the good whose consumption is positive, $x_l > 0$, that is, $\partial u(x^*) / \partial x_l / p_l = \lambda$. That is,

**Figure 2.8**

Corner solution where $x_1^* > 0$ but $x_2^* = 0$

$$\underbrace{\frac{\partial u(x^*)}{\partial x_l}}_{\substack{\text{MU per dollar} \\ \text{spent on good } l}} = \lambda \geq \underbrace{\frac{\partial u(x^*)}{\partial x_k}}_{\substack{\text{MU per dollar} \\ \text{spent on good } k}}.$$

Figure 2.8 illustrates a setting where the consumer seeks to consume good 1 alone, thus leading to a corner solution. Graphically, at the corner solution his indifference curve is steeper (more negative slope, $-MRS_{1,2}$) than his budget line, $-p_1/p_2$. That is, $-MRS_{1,2} < -p_1/p_2$, or $MRS_{1,2} > p_1/p_2$, which can be alternatively represented as $MU_1/MU_2 > p_1/p_2$, or $MU_1/p_1 > MU_2/p_2$. Intuitively, the individual would like to consume more of good 1, even after spending his entire wealth on good 1 alone.⁷

Lagrange multiplier The Lagrange multiplier is usually referred as the “marginal value of relaxing the constraint in the UMP,” or the “shadow price of wealth.” This

⁷ For more examples on Walrasian demands and their UMPs in the presence of interior and corner solutions, see exercises 1 and 2 at the end of the chapter.

interpretation originates from the fact that, if we relax the budget constraint in the UMP (i.e., if we provide more wealth to the consumer), he is capable of reaching a higher indifference curve and, as a consequence, obtaining a higher utility level. Figure 2.9 illustrates this point.

Hence we seek to measure what is the increase in utility resulting from a marginal increase in wealth. In order to do so, we take the individual's utility level measured at the utility maximizing bundle, $u(x(p, w))$. Differentiating $u(x(p, w))$ with respect to w , and using the chain rule, we obtain

$$\nabla u(x(p, w)) \cdot D_w x(p, w).$$

Substituting $\nabla u(x(p, w)) = \lambda p$ (one of our Kuhn–Tucker conditions for an interior solution in the UMP), we have

$$\lambda p \cdot D_w x(p, w).$$

From Walras's law, we know that $p \cdot x(p, w) = w$. Then the change in expenditure from an increase in wealth is given by

$$D_w [p \cdot x(p, w)] = p \cdot D_w x(p, w) = 1.$$

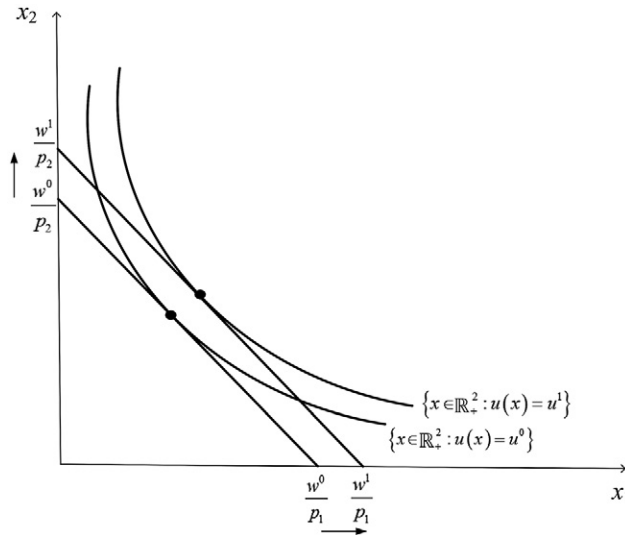


Figure 2.9
Interpretation of the Lagrange multiplier, λ

Hence

$$\nabla u(x(p, w)) \cdot D_w x(p, w) = \lambda \underbrace{p \cdot D_w x(p, w)}_1 = \lambda.$$

Since $p \cdot D_w x(p, w) = 1$, the marginal increase in wealth entails an increase in utility (at the optimum) of exactly λ .

2.2 Walrasian Demand—Comparative Statics

2.2.1 Wealth Changes

Let us now analyze how the Walrasian demand is affected by changes in the individual's wealth level. When demand increases in wealth, we say that good is a “normal” good, while when it decreases in wealth we refer to those goods as “inferior.” Examples of the former are computers and iPads. Examples of the latter are Two-Buck Chuck (a really cheap wine) and Walmart shopping during the economic crisis.⁸

Graphically, an increase in the wealth level produces an outward shift in the budget line, as figure 2.10 illustrates. In addition, at a given wealth level, the consumer chooses an optimal consumption bundle. We can connect all these optimal consumption

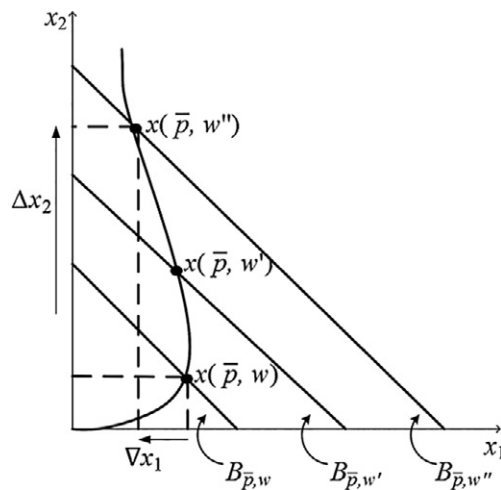


Figure 2.10
Wealth effects

⁸ Several reports suggest that a decrease in the average wealth during the 2009 economic crisis produced an increase in the sales of certain discount supermarkets, such as Costco, Walmart, and dollar stores.

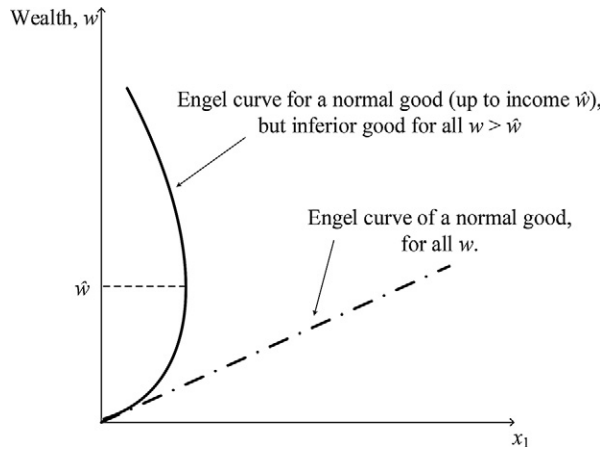


Figure 2.11
Engel curves

bundles for different levels of wealth, w , w' , and w'' , forming what we refer to as the “wealth expansion path.” When the wealth expansion path indicates an increase (decrease) in the consumption of good j as a consequence of further increments in the wealth level, we say that that good j is normal (inferior, respectively). Figure 2.10 illustrates an example in which x_1 , in the horizontal axis, is initially normal but becomes inferior, while x_2 is normal for all wealth levels.

We can alternatively measure how consumption is affected by wealth changes by using Engel curves, which depict the consumption of a particular good in the horizontal axis and wealth on the vertical axis. When the slope of the Engel curve is positive (negative), it indicates that the good is normal (inferior). Of course, an Engel curve can be positively sloped for low wealth levels but become negatively sloped afterward. Figure 2.11 illustrates an Engel curve for a normal good (positive slope for all wealth levels), and an Engel curve for a good whose demand is normal for relatively low wealth levels, $w < \hat{w}$, but becomes inferior beyond $\hat{w}$.

Example 2.2: Finding the wealth expansion path of a Cobb–Douglas utility function Consider a consumer with Cobb–Douglas utility function $u(x_1, x_2) = x_1^\alpha x_2^\beta$. In order to find the wealth-expansion path of this individual, we first need to identify his utility maximizing bundles, and then measure how these bundles are affected by wealth changes. In this context, the consumer’s UMP is

$$\begin{aligned} \max_{x_1, x_2} u(x_1, x_2) &= x_1^\alpha x_2^\beta \\ \text{subject to } p_1 x_1 + p_2 x_2 &\leq w. \end{aligned}$$

Taking first-order conditions with respect to x_1 yields

$$\alpha x_1^{\alpha-1} x_2^\beta - \lambda p_1 \leq 0,$$

where λ denotes the Lagrange multiplier. Similarly, taking first-order conditions with respect to x_2 , we find that

$$\beta x_1^\alpha x_2^{\beta-1} - \lambda p_2 \leq 0.$$

In the case of interior solutions, the preceding first-order conditions hold with equality. Solving for λ in both of them, we obtain

$$\frac{\alpha x_1^{\alpha-1} x_2^\beta}{p_1} = \frac{\beta x_1^\alpha x_2^{\beta-1}}{p_2}, \text{ or } x_2 = \frac{\beta}{\alpha} \frac{p_1}{p_2} x_1.$$

Hence this consumer's wealth expansion paths are described by the last equation, which is a ray from the origin with positive and constant slope $(\beta/\alpha)(p_1/p_2)$, implying that this individual regards both goods as normal (not inferior) in consumption. ■

2.2.2 Price Changes

Let us now analyze how demand reacts to price changes. When the demand for good k decreases as a result of an increase in the price of good k , we regard that good as a usual good, since its quantity demanded reacts negatively to its own price. If, in contrast, quantity demanded increases as a result of an increase in its own price, we regard it as a Giffen good.⁹ We illustrate these negative and positive relationships in figure 2.12, which represents the quantity demanded of good k in the horizontal axis and its own price in the vertical axis.

Besides analyzing the effect of a change in prices on its own quantity demanded (referred to as “own-price effects”), we are interested in examining the effect of a change in the price of good l on the quantity demanded for a different good k (referred to as “cross-price effects”). We can either find that this relationship is positive, when goods are regarded as substitutable (e.g., two brands of mineral water); or negative, for two goods regarded as complementary in consumption (e.g., cars and gasoline). We can use a similar graphical presentation as the one employed above in order to represent these cross-price effects (see figure 2.13).

In the left panel of figure 2.13, an increase in the price of one brand of mineral water, from p_l^0 to p_l^1 , increases the demand of the other brand of water (shifting its demand

⁹ One of the few examples of Giffen goods is that of potatoes in Ireland during the 19th century. However, there is still controversy among economists on whether demand for potatoes actually moved in the same direction as its own price.

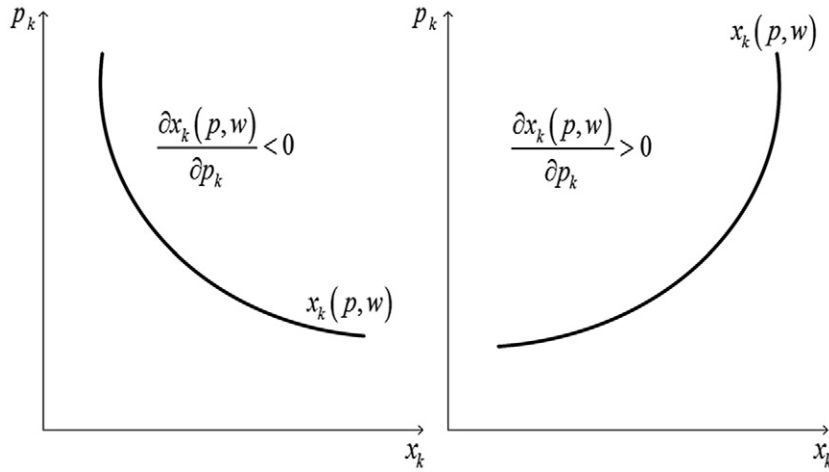


Figure 2.12
Own-price effects

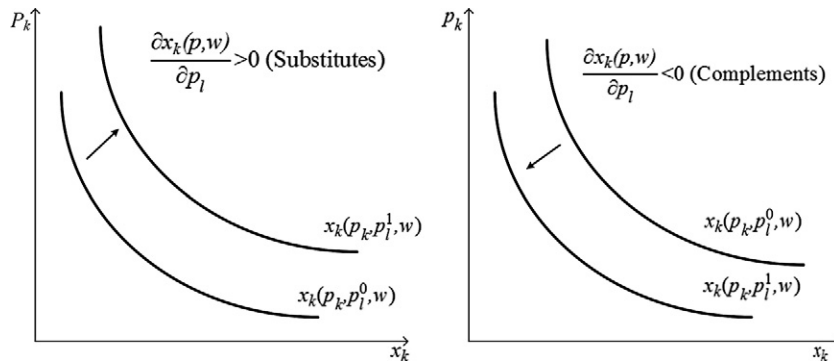


Figure 2.13
Cross-price effects for substitutes and complements.

curve outward), since the consumer regards them as close substitutes. Intuitively, the consumer reduces his demand of good l , as it became relatively more expensive, and increases his demand of good k . In the right panel, we observe how an increase in the price of gasoline from p_l^0 to p_l^1 reduces the demand for cars, shifting it inward. This case represents complementary goods.

2.2.3 Elasticities

When performing comparative statics analysis on demand, it is common to report the results in terms of elasticities. Let us next list some of the more typical elasticity measures on the Walrasian demand.

1. *Price elasticity* This elasticity measures the percentage change in the quantity demanded of good k that results from a 1 percent change in its price, p_k . That is,

$$\varepsilon_{x_k, p_k} = \frac{\% \Delta x_k(p, w)}{\% \Delta p_k} = \frac{\Delta x_k(p, w) / x_k(p, w)}{\Delta p_k / p_k} = \frac{\Delta x_k(p, w)}{\Delta p_k} \frac{p_k}{x_k(p, w)},$$

and in the case where the change in p_k is very small, the first term, $\Delta x_k(p, w) / \Delta p_k$, can be approximated by using the derivative. Hence

$$\varepsilon_{x_k, p_k} = \frac{\partial x_k(p, w)}{\partial p_k} \frac{p_k}{x_k(p, w)}.$$

For instance, in the case of the Walrasian demand that emerges from Cobb–Douglas preference over two goods $x_k(p, w) = \alpha_k w / p_k$, price elasticity is

$$\varepsilon_{x_k, p_k} = -\frac{\alpha_k w}{p_k^2} \frac{p_k}{\alpha_k w / p_k} = -1.$$

2. *Cross-price elasticity* This elasticity also considers a 1 percent increase in the price of good k but measures the percent change in the Walrasian demand of the other good $j \neq k$. That is,

$$\varepsilon_{x_k, p_j} = \frac{\partial x_k(p, w)}{\partial p_j} \frac{p_j}{x_k(p, w)}.$$

Following with our Cobb–Douglas example, where $x_k(p, w) = \alpha_k w / p_k$, cross-price elasticity is zero, $\varepsilon_{x_k, p_j} = 0$, since the Walrasian demand of good k is unaffected by changes in the price of good j , that is, $\partial x_k(p, w) / \partial p_j = 0$.

3. *Income elasticity* This elasticity measures how the Walrasian demand for a good, $x_k(p, w)$, is affected by a 1 percent change in income, w . That is,

$$\varepsilon_{x_k, w} = \frac{\partial x_k(p, w)}{\partial w} \frac{w}{x_k(p, w)}.$$

In the case of the Cobb–Douglas utility, since $x_k(p, w) = \alpha_k w / p_k$, the income elasticity becomes

$$\varepsilon_{x_k, w} = \frac{\alpha_k}{p_k} \frac{w}{\alpha_k w / p_k} = 1.$$

2.3 Indirect Utility Function

Besides the set of properties of the optimal consumption bundle that solves the UMP (Walrasian demand), there are some important points about the UMP that we should consider. First, when we insert the optimal consumption bundle into the individual's utility function, we obtain the highest utility level that the individual can achieve by solving his UMP, $u(x(p, w))$, or compactly referred to as the indirect utility function, and denoted by $v(p, w)$.¹⁰ Function $v(p, w)$ satisfies several properties:

Homogeneity of Degree Zero

This property states that increasing market prices and wealth by the same proportion does not modify the consumer's budget set (see figure 2.14). As a consequence, such an increase does not modify the consumer's optimal consumption bundle, $x(p, w)$, nor does it affect the maximal utility level that the individual reaches in his UMP, as measured by $v(p, w)$, meaning $v(\alpha p, \alpha w) = v(p, w)$.

Strictly Increasing in w , and Nonincreasing in p_k for Every Good k

This property states that if we increase the wealth of the individual, we enlarge the set of feasible bundles he can afford and, as a consequence, we allow him to reach an indifference curve associated with a higher utility level. Therefore the maximal utility level that he can obtain is strictly increasing in his wealth, as depicted in figure 2.15 where $v(p, w') > v(p, w)$ because $w' > w$.

In contrast, with an increase in the price of any good, the set of affordable bundles shrinks, so the individual can only reach indifference curves associated with weakly lower utility levels. Hence, an increase in the price of any good k produces a weak reduction in the maximal utility level that the individual can obtain by solving this UMP (see figure 2.16).

Remark By the property above, $v(p, w)$ is nonincreasing in p_k , thus allowing $v(p, w)$ to remain unaffected as the consumer experiences an increase in the price of the good. This can occur when he is initially at a corner solution, and a change in p_k does not modify his optimal consumption bundle, thus not affecting $v(p, w)$ either. Nonetheless, if p_k changes sufficiently, even a consumer who is initially at a corner solution will

¹⁰ More generally, we refer to the objective function of an optimization problem evaluated at the solution of such optimization problem as the "value function." Hence, the value function of the UMP is the indirect utility function.

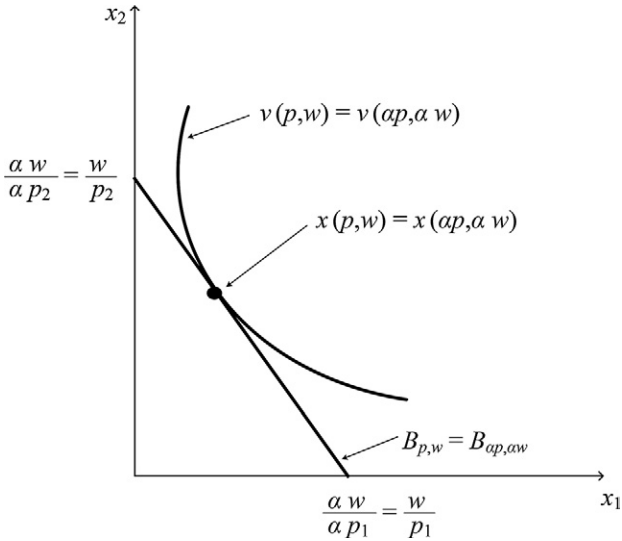


Figure 2.14
 $v(p, w)$ homogeneous of degree zero

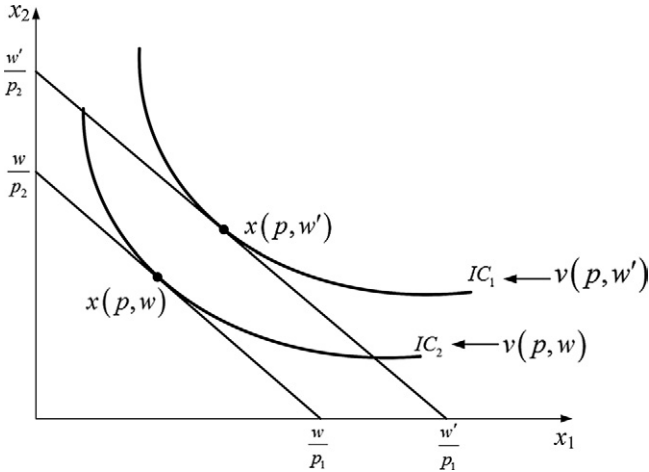


Figure 2.15
 $v(p, w)$ strictly increasing in w

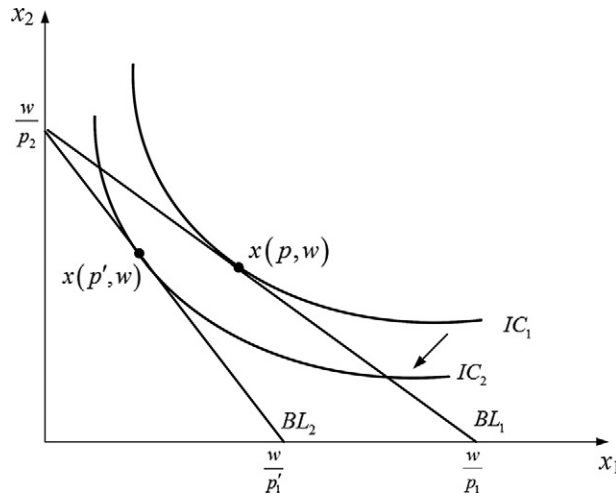


Figure 2.16

$v(p, w)$ is nonincreasing in p_k

change his optimal consumption bundle by potentially moving to a different corner solution, ultimately lowering his utility level as measured by $v(p, w)$.

Quasi-Convexity

The set $\{(p, w) : v(p, w) \leq \bar{v}\}$ is convex for any $\bar{v} \in \mathbb{R}_+$. An intuitive explanation of this property is provided in figure 2.17. First, the figure depicts the indirect utility function with the price of good 1 on the horizontal axis and the wealth level on the vertical axis. Hence, when the price of good 1 increases from p_1^1 to p_1^2 , this individual's wealth must also increase in order to keep his utility level unaffected. In contrast, lower prices and higher wealth levels are associated with higher maximal utilities. Graphically, moving to the northwest in figure 2.17 entails a higher utility level, $v(p^*, w^*) > v(p, w)$, where $p^* < p$ and $w^* > w$.

Let us now examine the quasi-convexity of $v(p, w)$. Specifically, this property implies that, if the maximum utility associated with a given pair of prices and wealth (point A in figure 2.17) is weakly higher than the maximum utility associated with another pair of prices and wealth (point B), then the convex combination of points A and B (point C) yields a maximum utility that is weakly lower than that associated with A . That is, if the consumer faces an intermediate price level $\lambda p_1^1 + (1 - \lambda)p_1^2$ and an

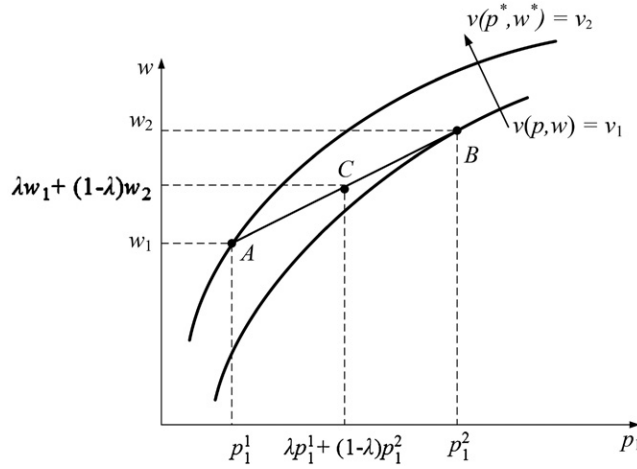


Figure 2.17
Quasi-convexity of $v(p, w)$ —I

intermediate wealth level $\lambda w_1 + (1 - \lambda)w_2$, the maximum utility level that he will be able to reach by solving his UMP is *lower* than that at point $A \equiv (p_1^1, w_1)$.

An alternative interpretation of quasi-convexity simply states that the pairs of prices and wealth for which the consumer reaches a lower utility level than that under pair (p^*, w^*) defines a convex set. More compactly, $v(p, w)$ is quasi-convex if the set of (p, w) pairs for which $v(p, w) < v(p^*, w^*)$ is convex, meaning $\{(p, w) : v(p, w) < v(p^*, w^*)\}$ is convex. Figure 2.18 illustrates this property by first depicting the utility level associated with the price-wealth pair (p^*, w^*) , $v(p^*, w^*)$, and then shading the set of price-wealth pairs that yield a lower utility level than $v(p^*, w^*)$ does, thus yielding a convex set.

Another common illustration of quasi-convexity uses goods 1 and 2 on the axis (see figure 2.19). First, when the individual faces budget set $B_{p,w}$, his optimal consumption bundle is $x(p, w)$. Second, when prices and wealth change to p' and w' , he faces budget set $B_{p',w'}$, and selects bundle $x(p', w')$. Third, both bundles $x(p, w)$ and $x(p', w')$ induce the same utility $\bar{u}$ because they lie on the same indifference curve, so the indirect utility function satisfies $v(p, w) = v(p', w') = \bar{u}$. We can now construct a convex combination of prices and wealth:

$$p'' = \alpha p + (1 - \alpha)p' \quad \text{and} \quad w'' = \alpha w + (1 - \alpha)w',$$

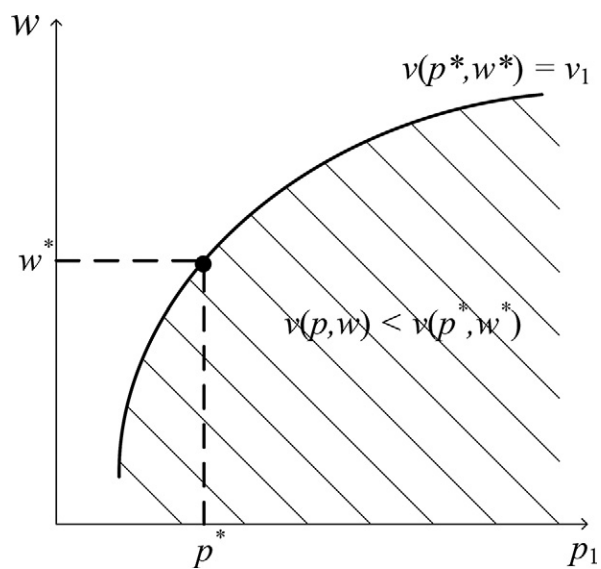


Figure 2.18
Quasi-convexity of $v(p, w)$ —II

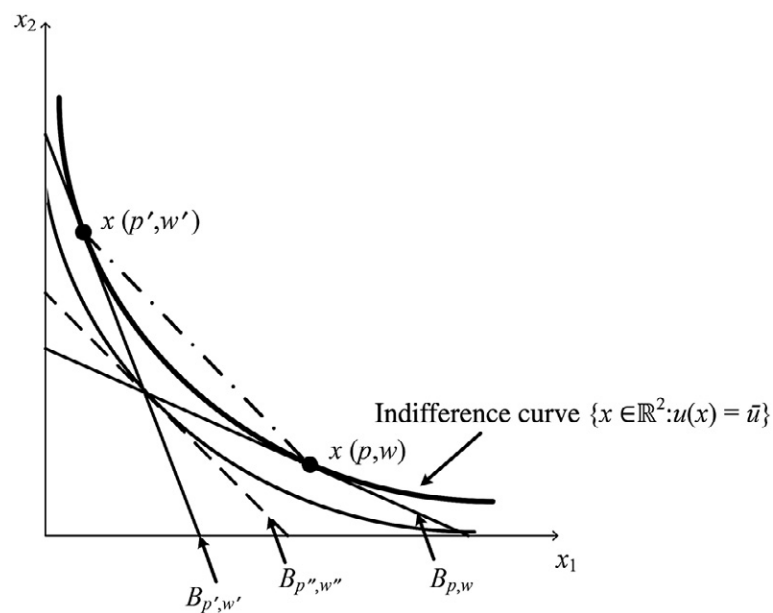


Figure 2.19
Quasi-convexity of $v(p, w)$ —III

which provides us with budget set $B_{p'', w''}$ (shown with dashed lines in figure 2.19). Finally, note that any solution to the UMP when the consumer faces budget set $B_{p'', w''}$ must provide an optimal consumption bundle that lies on a lower indifference curve (associated with a lower utility level) than $\bar{u}$. Hence the combination of two price-wealth pairs must yield a utility level lower than that from either of the price-wealth pairs alone.

Example 2.3: $v(p, w)$ associated with the Cobb–Douglas utility function Consider a consumer with Cobb–Douglas utility function $u(x_1, x_2) = x_1^\alpha x_2^\beta$. From the previous section, we know that the Walrasian demands for good 1 and 2 are

$$x_1(p, w) = \frac{\alpha w}{(\alpha + \beta)p_1}, \quad x_2(p, w) = \frac{\beta w}{(\alpha + \beta)p_2}.$$

As a consequence the indirect utility function is

$$v(p, w) = (x_1(p, w))^\alpha (x_2(p, w))^\beta = \left(\frac{\alpha w}{(\alpha + \beta)p_1} \right)^\alpha \left(\frac{\beta w}{(\alpha + \beta)p_2} \right)^\beta,$$

which reduces to

$$v(p, w) = \left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{\beta}{p_2} \right)^\beta \left(\frac{w}{\alpha + \beta} \right)^{\alpha + \beta}.$$

As expected, this indirect utility function is decreasing in the price of both goods, increasing in the consumer's wealth, w , and homogeneous of degree zero, since increasing all prices and wealth by a common factor α yields

$$\begin{aligned} v(\alpha p, \alpha w) &= \left(\frac{\alpha}{\alpha p_1} \right)^\alpha \left(\frac{\beta}{\alpha p_2} \right)^\beta \left(\frac{\alpha w}{\alpha + \beta} \right)^{\alpha + \beta} = (\alpha)^{-\alpha - \beta + \alpha + \beta} \left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{\beta}{p_2} \right)^\beta \left(\frac{w}{\alpha + \beta} \right)^{\alpha + \beta} \\ &= \left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{\beta}{p_2} \right)^\beta \left(\frac{w}{\alpha + \beta} \right)^{\alpha + \beta} = v(p, w), \end{aligned}$$

as required for homogeneity of degree zero. ■

2.4 WARP and Demand

Having explored the different properties of the UMP, its solution, $x(p, w)$, and its value function, $v(p, w)$, we can now connect the optimal consumption bundle and the WARP. In particular, we want to learn if the consistency requirement imposed by WARP limits

the set of optimal consumption bundles that the individual decision maker can select when solving the UMP.

Consider two different consumption bundles $x(p, w)$ and $x(p', w')$, both being affordable under (p, w) , that is, $p \cdot x(p, w) \leq w$ and $p \cdot x(p', w') \leq w$. When prices and wealth are (p, w) , the consumer chooses $x(p, w)$ despite $x(p', w')$ also being affordable. Therefore the consumer “reveals” a preference for $x(p, w)$ over $x(p', w')$, though both are affordable (consistency). However, bundle $x(p, w)$ may not be affordable at (p', w') , since the consumer chose $x(p', w')$. That is, $p' \cdot x(p, w) > w'$. We can summarize these properties as follows:

WARP Walrasian demand satisfies WARP if, for two different consumption bundles, $x(p, w) \neq x(p', w')$:

$$p \cdot x(p', w') \leq w \Rightarrow p' \cdot x(p, w) > w'.$$

In words, if bundle $x(p, w)$ and $x(p', w')$ are affordable under budget set $B_{p, w}$, and the consumer selects $x(p, w)$ rather than $x(p', w')$, then bundle $x(p, w)$ must not be affordable under budget set $B_{p', w'}$, since the consumer selected bundle $x(p', w')$.

Let us begin by presenting optimal consumption bundles that satisfy WARP. In figure 2.20, bundles $x(p, w)$ and $x(p', w')$ are both affordable under (p, w) , since they both lie on or below budget line $B_{p, w}$. In contrast, bundle $x(p, w)$ is not affordable under (p', w') because it lies above budget line $B_{p', w'}$. Therefore WARP is satisfied.¹¹

Let us next examine the case in figure 2.21, where demand under final prices and wealth, $x(p', w')$, is not affordable under initial prices and wealth because it lies above budget line $B_{p, w}$. In this case we cannot check if WARP applies, since the premise of WARP is not satisfied in the first place given that bundle $x(p', w')$ is not affordable under (p, w) . Hence WARP is not violated. A similar argument applies to the bundles in figure 2.22, since $x(p', w')$ is not affordable under initial prices and wealth (p, w) .

2.4.1 Testing WARP

The procedure that we have been using to test whether two particular bundles satisfy WARP can be sequentially described as follows:

¹¹ Note that we first checked whether the premise of WARP holds, that is, whether both bundles $x(p, w)$ and $x(p', w')$ are affordable under (p, w) , thus lying on or below budget line $B_{p, w}$. We then checked whether the conclusion of WARP is satisfied, that is, whether bundle $x(p, w)$ is unaffordable under (p', w') , thus lying above budget line $B_{p', w'}$. In this example, the premise was satisfied, as well as its conclusion.

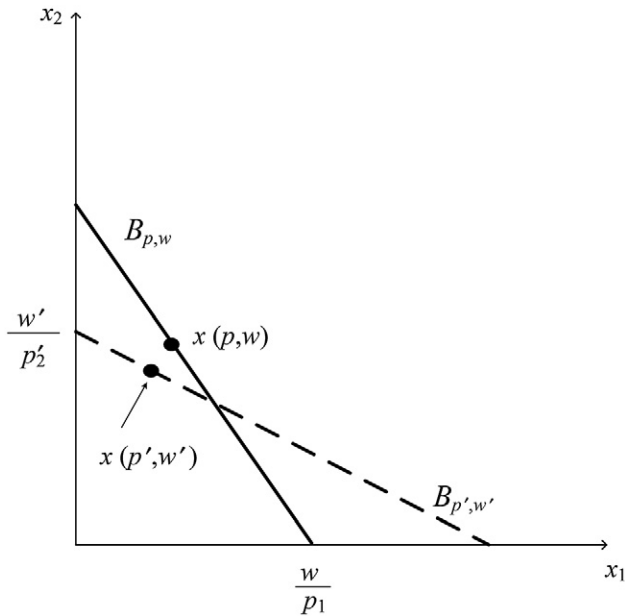


Figure 2.20
Walrasian demand that satisfies WARP

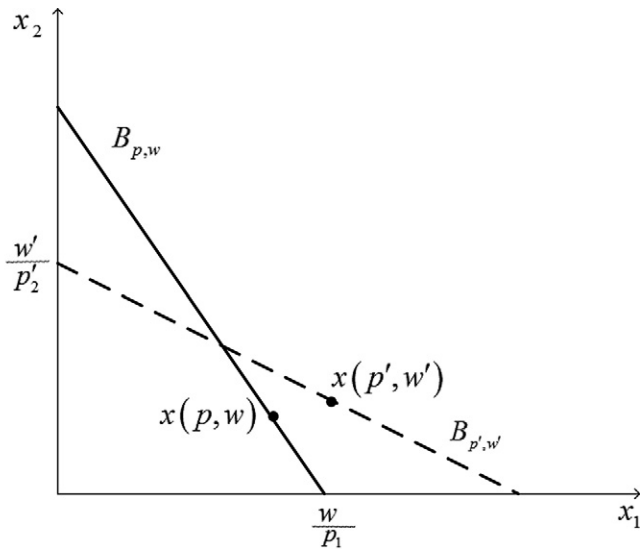


Figure 2.21
Walrasian demand that does not violate WARP

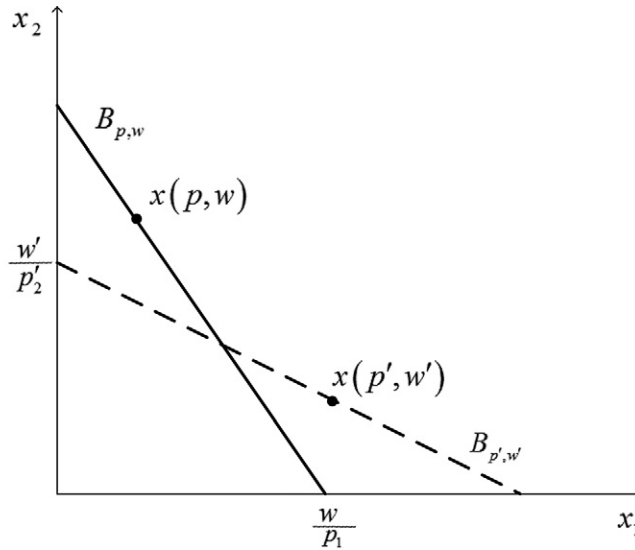


Figure 2.22
Walrasian demand that does not violate WARP

1. We check if bundles $x(p, w)$ and $x(p', w')$ are both affordable under the initial prices and wealth. Graphically, both bundles must lie on or below budget set $B_{p,w}$:
 - If step 1 is satisfied, then we can move to step 2.
 - Otherwise, the premise of WARP is not satisfied, not allowing us to continue checking whether it is violated. In such cases we say that the WARP is “not violated.” (Recall that for us to be able to claim that WARP is violated, we must find that its premise holds, namely that $p \cdot x(p', w') \leq w$, but its conclusion does not, that is $p' \cdot x(p, w) < w'$.)
2. We check if bundle $x(p, w)$ is *not* affordable under final prices and wealth. Graphically, bundle $x(p, w)$ must lie on or above budget line $B_{p',w'}$.
 - If this condition is satisfied, then the Walrasian demand satisfies WARP.
 - If, in contrast, this condition does not hold, then the Walrasian demand violates WARP.

Figure 2.23 represents a similar case to those previously analyzed. Bundle $x(p', w')$ is not affordable under the price-wealth pair (p, w) because $x(p', w')$ lies above $B_{p,w}$. Hence the premise of WARP does not hold, which, as in our previous examples, means that WARP is not violated.

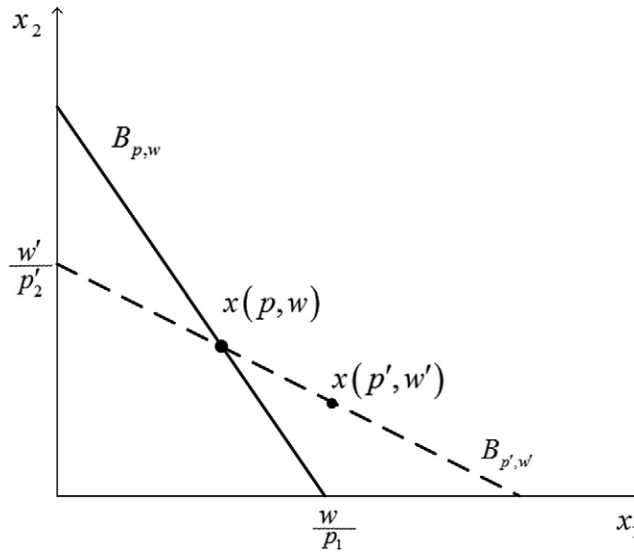


Figure 2.23
WARP not violated

Suppose that, as shown in figure 2.24, the optimal consumption bundle under final prices and wealth, $x(p', w')$, is affordable at initial prices and wealth, since it lies below the budget line $B_{p,w}$. Similarly bundle $x(p, w)$ is affordable under $B_{p',w'}$, and hence the premise of WARP is satisfied. Applying the second step of the WARP procedure, we can see that the optimal consumption bundle $x(p, w)$ is unaffordable under the new prices and wealth, given that it lies above budget line $B_{p',w'}$. Hence WARP is satisfied.

Our last example in figure 2.25 exhibits a similar pattern. Specifically, the optimal consumption bundle with final prices and wealth, $x(p', w')$, is affordable under (p, w) , since it lies below budget line $B_{p,w}$. A similar argument applies to bundle $x(p, w)$, which lies on $B_{p,w}$, thus being affordable as well, so the premise of WARP holds. Furthermore, upon applying the second step of the WARP procedure, we learn that although the demand $x(p, w)$ is affordable under the new prices and wealth, it lies below budget set $B_{p',w'}$. Therefore WARP is not satisfied.¹²

2.4.2 Implications of WARP

WARP has some important implications for the set of admissible consumption bundles that a consumer may choose before and after a price change. Let us begin by

¹² For more examples on demand functions satisfying or violating WARP, see exercises 5 through 7 at the end of this chapter.

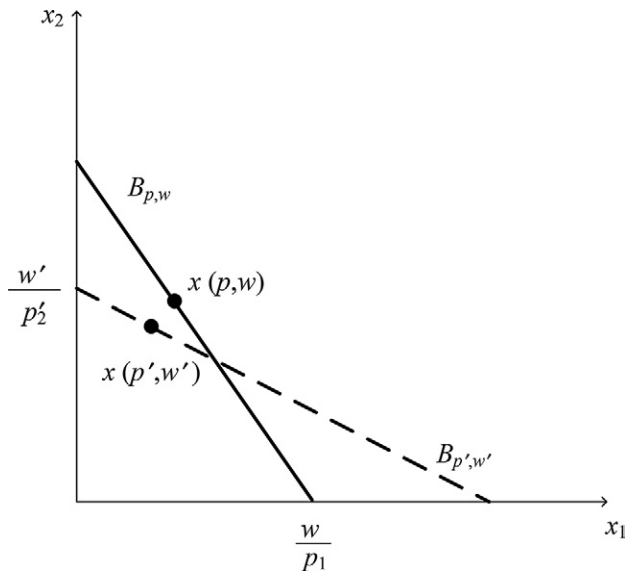


Figure 2.24
WARP satisfied

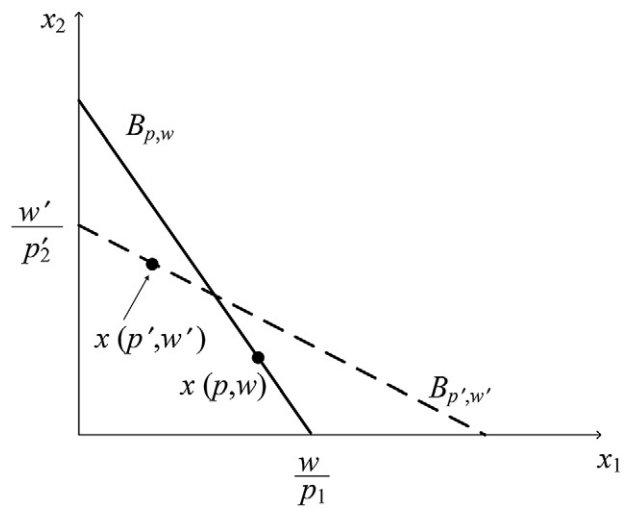


Figure 2.25
WARP not satisfied

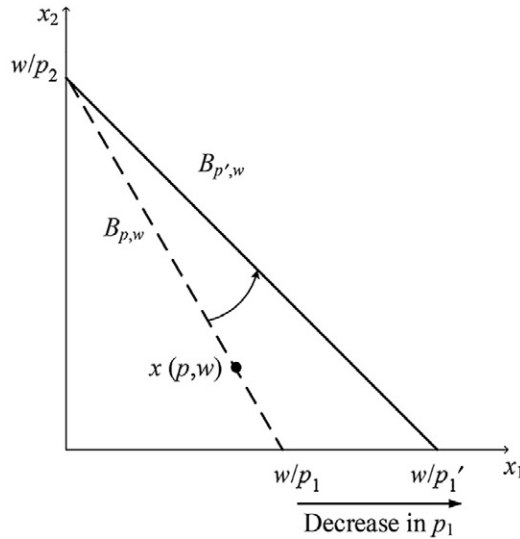


Figure 2.26
Uncompensated price change

considering a reduction in the price of good 1, as figure 2.26 illustrates with an upward pivot on the budget line.

After the price change, we adjust the consumer's wealth so that he can consume his initial demand $x(p, w)$ at the new (lower) prices. Therefore we shift the final budget line $B_{p',w}$ inward (reducing this consumer's wealth) until the point at which we reach the initial consumption bundle $x(p, w)$. Importantly, the budget line after the shift (after the reduction in wealth) is parallel to budget line $B_{p',w}$, reflecting the final price ratio.

What is the precise reduction in wealth that we must apply to this consumer in order for him to exactly afford bundle $x(p, w)$? To find this wealth adjustment, the individual must spend $p \cdot x(p, w) = w$ in order to buy bundle $x(p, w)$ at price p but must spend $p' \cdot x(p, w) = w'$ when the market price is p' . The difference in his expenditure is

$$w' - w = p' \cdot x(p, w) - p \cdot x(p, w),$$

$$\text{or } \Delta w = \Delta p \cdot x(p, w),$$

where $\Delta w = w' - w$ and $\Delta p = p' - p$. This wealth adjustment is the Slutsky wealth compensation. Intuitively, it represents the increase (decrease) in wealth, measured by Δw , that must be provided to the consumer in order for him to afford the original bundle $x(p, w)$ after experiencing a price increase (decrease, respectively). In figure 2.27 the

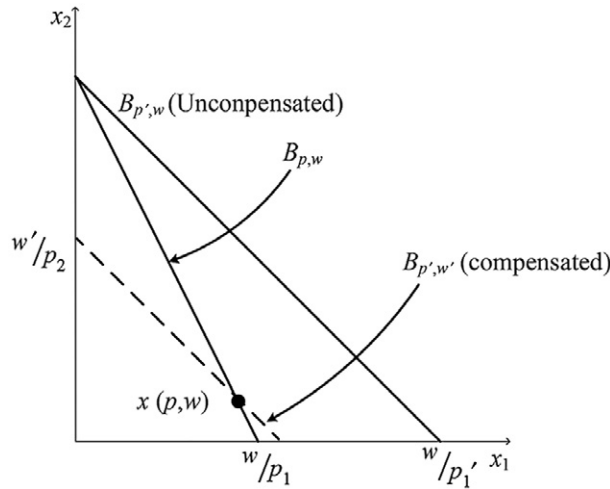


Figure 2.27
Uncompensated and compensated price change

Slutsky wealth compensation is illustrated by the reduction in the consumer's wealth that affords him the same consumption bundle as before the price decrease.¹³ Given this interpretation of Slutsky wealth compensation, we are ready to establish a relationship between the law of demand and WARP. Suppose that the Walrasian demand $x(p, w)$ satisfies homogeneity of degree zero and Walras's law. Then $x(p, w)$ satisfies WARP if and only if $\Delta p \cdot \Delta x \leq 0$, where $\Delta x = x(p', w') - x(p, w)$ and where $x(p', w')$ is evaluated at w' , which is the wealth level that allows the consumer to buy his initial demand at the new prices, $w' = p' \cdot x(p, w)$

This result establishes that if, after the price change, the consumer's wealth is compensated "à la Slutsky," as described above, then the WARP becomes equivalent to the law of demand (i.e., quantity demanded and price move in different directions). That is, in the case of a price decrease, the WARP implies that quantity demanded must increase, provided that the consumer's wealth is adjusted "à la Slutsky."

Figure 2.28 depicts another example in which the WARP restricts behavior when the Slutsky wealth compensation is applied. In the figure the price change is similar to that

¹³ In contrast, the so-called Hicksian wealth compensation adjusts the wealth level of the individual after the price change, so that he can still reach the same indifference curve (utility level) he was reaching before the price change (rather than reaching the same consumption bundle, as in the Slutsky consumption). We comment on this type of wealth compensation in subsequent sections of this chapter.

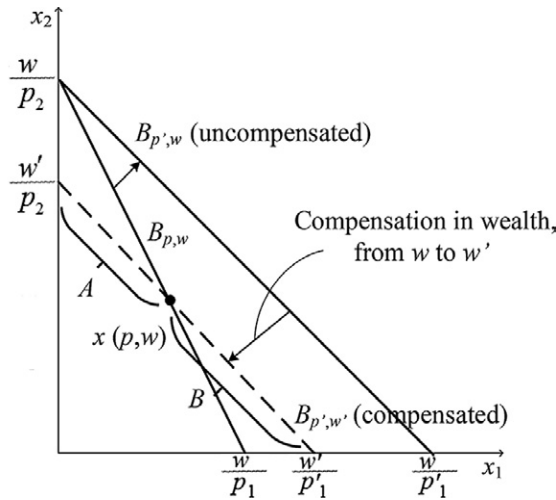


Figure 2.28
Compensated law of demand and WARP

represented above, where the price of x_2 is not affected but the price of x_1 decreases. By pivoting the budget line inward, we present the consumer with Slutsky wealth compensation so that he can still afford the initial bundle $x(p, w)$. The consumer's budget line after the wealth compensation is hence $B_{p',w'}$, as depicted by the dashed line. A natural question at this point is: Where does bundle $x(p', w')$ lie? Let us first examine whether such a bundle might lie to the left-hand side of bundle $x(p, w)$, that is, on segment A in figure 2.28. The first step of WARP is satisfied because both bundles $x(p, w)$ and $x(p', w')$ are affordable under budget set $B_{p,w}$, since they both lie on or below $B_{p,w}$. When we test the second step of WARP, however, we find that bundle $x(p, w)$ is affordable under final prices and wealth, given that it lies on budget set $B_{p',w'}$, in violation of WARP. Therefore bundle $x(p', w')$ cannot lie on segment A .

Let us now examine whether bundle $x(p', w')$ can lie to the right-hand side of bundle $x(p, w)$, on segment B . Bundle $x(p, w)$ is affordable under initial prices and wealth, since it lies on budget set $B_{p,w}$, but bundle $x(p', w')$ would not be affordable, given that it would lie above budget set $B_{p,w}$. Hence WARP's premise does not hold. As a consequence WARP would not be violated if bundle $x(p', w')$ lies on segment B . The fact that bundle $x(p', w')$ can lie on segment B implies that bundle $x(p', w')$ must contain more of x_1 than bundle $x(p, w)$ does. We can therefore conclude that a decrease in the price of x_1 (when we appropriately compensate wealth effects) leads to an increase in the quantity

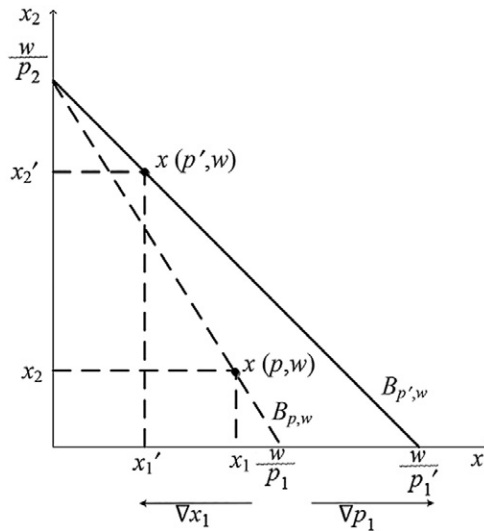


Figure 2.29
Uncompensated price change

demanded for such a good. This is what we refer as the *compensated* law of demand (CLD).¹⁴

Note an important distinction between the *uncompensated* law of demand and the CLD just described. Specifically, the demand for x_1 can fall as a consequence of a decrease in its price (i.e., quantity demanded and price can move in the same direction) only when wealth is left uncompensated, as in figure 2.29.

The figure shows a reduction in the price of x_1 similar to what we analyzed before. Individual demand after the price change is given by $x(p', w)$ so that the quantity demanded of x_1 decreases even after the good became cheaper. In this case the uncompensated law of demand is not satisfied because quantity demanded and price move in the same direction. This is why WARP is not a sufficient condition to yield the uncompensated law of demand. Clearly, WARP and the CLD are equivalent, but WARP and the uncompensated law are not necessarily related. We can examine the last point by checking whether the WARP was satisfied under the uncompensated law of demand. In particular, bundle $x(p, w)$ was affordable under the initial budget set $B_{p,w}$, but the

¹⁴ *Practice:* Can you repeat this analysis for the case of an increase in the price of x_1 ? First, you need to pivot the budget line inwards. Second, note that the wealth compensation must imply in this case an increase in consumer's wealth. Finally, you will have a budget set after the wealth compensation with two segments A and B . Determine which one is allowed/not allowed according to WARP.

consumption bundle after the (uncompensated) price change $x(p', w')$ was not affordable, since it lies above budget set $B_{p, w}$. Therefore the premise of WARP is not satisfied and WARP is not violated. The example in figure 2.29 shows a case in which WARP is not violated but the uncompensated law of demand does not hold. As a consequence the WARP and the uncompensated law of demand are not necessarily satisfied at the same time. (For more on the relationship between CLD and WARP, see exercise 11 at the end of this chapter.)

2.5 Slutsky Matrix

The Walrasian demand function is differentiable in both prices and wealth under relatively general conditions. We next further examine the relationship between the CLD and the WARP. In order to do so, let us first totally differentiate the Walrasian demand function, as follows:

$$dx = D_p x(p, w)dp + D_w x(p, w)dw.$$

And since the consumer's wealth is compensated, $dw = x(p, w) \cdot dp$ (this is the differential analogue of $\Delta w = \Delta p \cdot x(p, w)$ in the Slutsky wealth compensation described in the previous section). Substituting $dw = x(p, w) \cdot dp$ in the expression above obtains

$$dx = D_p x(p, w)dp + D_w x(p, w) \underbrace{[x(p, w) \cdot dp]}_{dw},$$

or equivalently,

$$dx = [D_p x(p, w) + D_w x(p, w)x(p, w)^T]dp,$$

where the subscript T denotes the transposed vector $x(p, w)$. Hence the CLD, $dp \cdot dx \leq 0$, can be also expressed as

$$dp \cdot [D_p x(p, w) + D_w x(p, w)x(p, w)^T]dp \leq 0,$$

where the term in brackets is the so-called Slutsky (or substitution) matrix, $S(p, w)$. In particular,

$$S(p, w) = \begin{bmatrix} s_{11}(p, w) & \dots & s_{1N}(p, w) \\ \vdots & \ddots & \vdots \\ s_{N1}(p, w) & \dots & s_{NN}(p, w) \end{bmatrix},$$

where each term of the matrix is

$$s_{lk}(p, w) = \frac{\partial x_l(p, w)}{\partial p_k} + \frac{\partial x_l(p, w)}{\partial w} x_k(p, w).$$

for every two goods l and k .

We next examine under which conditions the own-price effects, as represented by the elements in the main diagonal of the matrix, $s_{ll}(p, w)$, are negative. If $s_{ll}(p, w) \leq 0$ for all goods, then the Slutsky matrix would be negative semidefinite. In particular, if $x(p, w)$ is differentiable, satisfies Walras's law, homogeneity of degree zero and WARP, then the Slutsky matrix $S(p, w)$ is negative semidefinite. That is, $v \cdot S(p, w) \cdot v \leq 0$ for any vector $v \in \mathbb{R}^N$ and all terms in the main diagonal of the matrix must be weakly negative, meaning own-price effects satisfy $s_{ll}(p, w) \leq 0$.¹⁵ As described above, the substitution effect $s_{ll}(p, w)$ embodies two effects:

$$\underbrace{s_{ll}(p, w)}_{\text{Substitution effect } (-)} = \underbrace{\frac{\partial x_l(p, w)}{\partial p_l}}_{\substack{\text{Total effect} \\ (-) \text{ for usual goods} \\ (+) \text{ for Giffen goods}}} + \underbrace{\frac{\partial x_l(p, w)}{\partial w} x_l(p, w)}_{\substack{\text{Income effect} \\ (+) \text{ for normal goods} \\ (-) \text{ for inferior goods}}}.$$

We can rearrange this expression to state that the total effect is equal to the substitution effect minus the income effect, $TE = SE - IE$.¹⁶ Let us describe each of the terms in the expression above. First, from our previous discussion on Slutsky compensation and the CLD, we know that the substitution effect is negative for all types of goods because $s_{ll}(p, w) \leq 0$. Second, the total effect measures the change in the quantity demanded for good l as a result of a change in the price of good l . Hence it considers an uncompensated price effect, given that it reflects a change in the price of good l without adjusting the wealth of the individual. Third, the income effect describes the change in demand due to the adjustment in wealth as a result of a Slutsky wealth compensation.¹⁷

¹⁵ The fact that the substitution matrix is negative semidefinite does not imply that this matrix is symmetric. Nevertheless if the utility function is continuous and represents a strictly convex preference relation that satisfies LNS, then the substitution matrix is symmetric. In many applications, these conditions hold and therefore the substitution matrix is symmetric. However, in the case of preferences for perfect substitutes, indifference curves do not satisfy strict convexity and, hence, we cannot use the above result to guarantee that the substitution matrix is symmetric.

¹⁶ Recall that when the quantity demanded of good l decreases (increases) in the price of good l , we refer to that good as usual (Giffen, respectively). Similarly, when the quantity demanded of good l increases (decreases) in wealth, we refer to that good as normal (inferior, respectively).

¹⁷ Recall the difference between a Slutsky and a Hicksian wealth compensation. The former adjusts the consumer's wealth so that he can still afford the same bundle that he purchased before the price change, whereas the latter adjusts wealth so that the consumer can still reach the same utility level as before the price change (graphically reaching the same indifference curve).

Figure 2.30 presents graphically the intuition to a reduction of the price of good 1, where such a good is considered “normal.” First, the reduction in the price of x_1 enlarges the set of feasible bundles for the consumer, allowing him to reach an indifference curve farther from the origin, at which he increases the amount of x_1 consumed from x_1^0 to x_1^2 . A graphical representation of the associated Walrasian demand is depicted in the lower panel. The demand curve indicates that a decrease in the price of x_1 leads to an increase in the quantity demanded.

This increase in the quantity demanded represents the total effect. Nonetheless, it is interesting to separate the total effect into two components: the substitution and income effect. Figure 2.31 replicates the reduction in the price of x_1 and the total effect but also includes the Slutsky wealth compensation. In figure 2.31, after the price change, we reduce this consumer’s wealth so that he can afford the same consumption bundle that he was buying before the price change, A . Graphically we do so after the price change by shifting the budget line inward until it “crosses” through the initial bundle A . The consumer can afford any consumption bundle along the new budget line. In the top panel of figure 2.31, the consumer’s optimal bundle involves buying only x_1^3 units. The bottom panel of figure 2.31 replicates the Walrasian demand, but we also include the so-called constant purchasing power demand curve (CPP curve), resulting from applying the Slutsky wealth compensation. Specifically, a given reduction in the price of x_1 produces a relatively small increase in the quantity demanded for x_1 when we hold the consumer’s purchasing power constant (applying the Slutsky wealth compensation), thus only increasing his consumption from x_1^0 to x_1^3 (from point a to c). However, this price reduction implies a relatively large increase in the quantity demanded for x_1 when we do not hold the consumer’s purchasing power constant, as measured by the Walrasian demand, since his consumption increases from x_1^0 to x_1^2 (from point a to b).

In figure 2.32 we replicate figure 2.31, adding now the so-called Hicksian demand (also referred to as “constant utility” demand curve). The top panel of figure 2.32 shows the Hicksian wealth compensation after the price change. The consumer’s wealth level is adjusted so that he can still reach his initial utility level (graphically reaching the same indifference curve I_1 as before the price change). In this graphical example, we observe a more significant wealth reduction than when we apply the Slutsky wealth compensation. Therefore the new budget line (after the wealth compensation) is tangent to the initial indifference curve at a point where the consumer demands x_1^1 units of good one, which is bundle D . In the bottom panel of figure 2.32, the Hicksian demand curve reflects that, for a given decrease in p_1 , the consumer slightly increases his consumption of good 1 from x_1^0 to x_1^1 (from point a to d). However, such

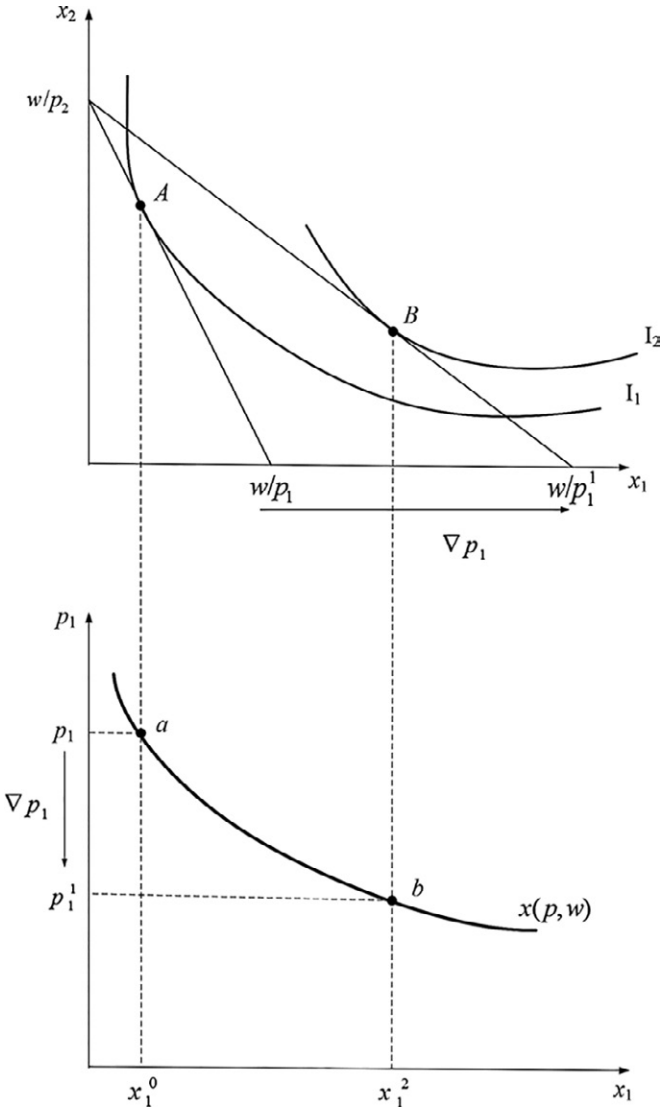


Figure 2.30
Constructing Walrasian demand: Total effect

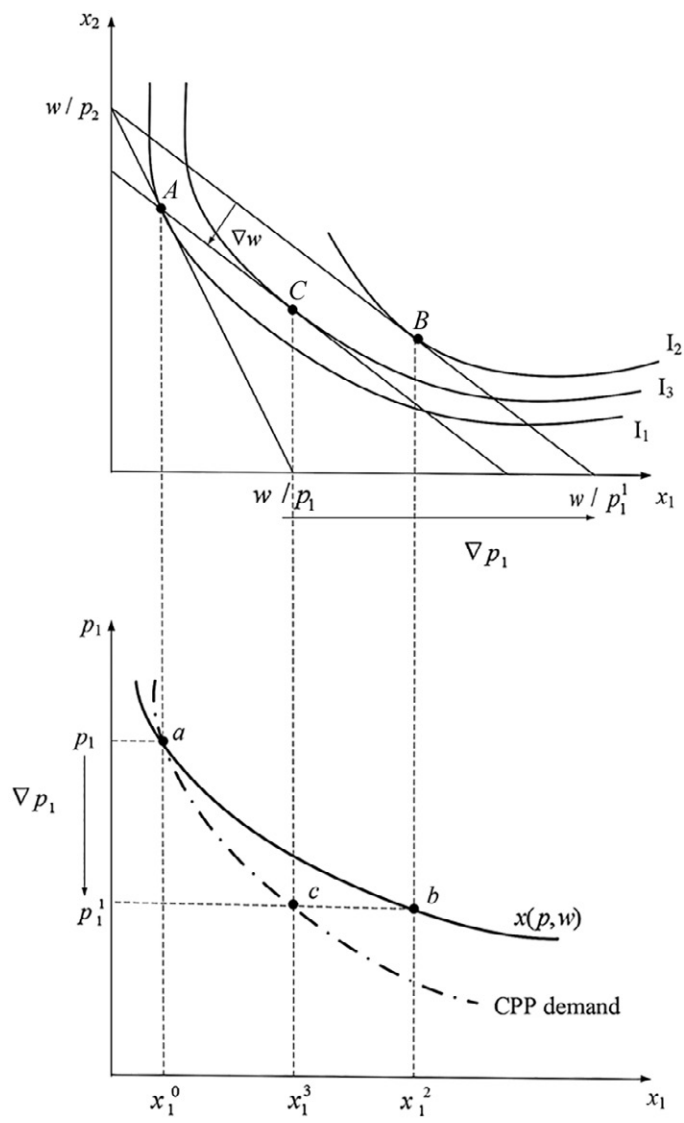


Figure 2.31
Slutsky wealth compensation

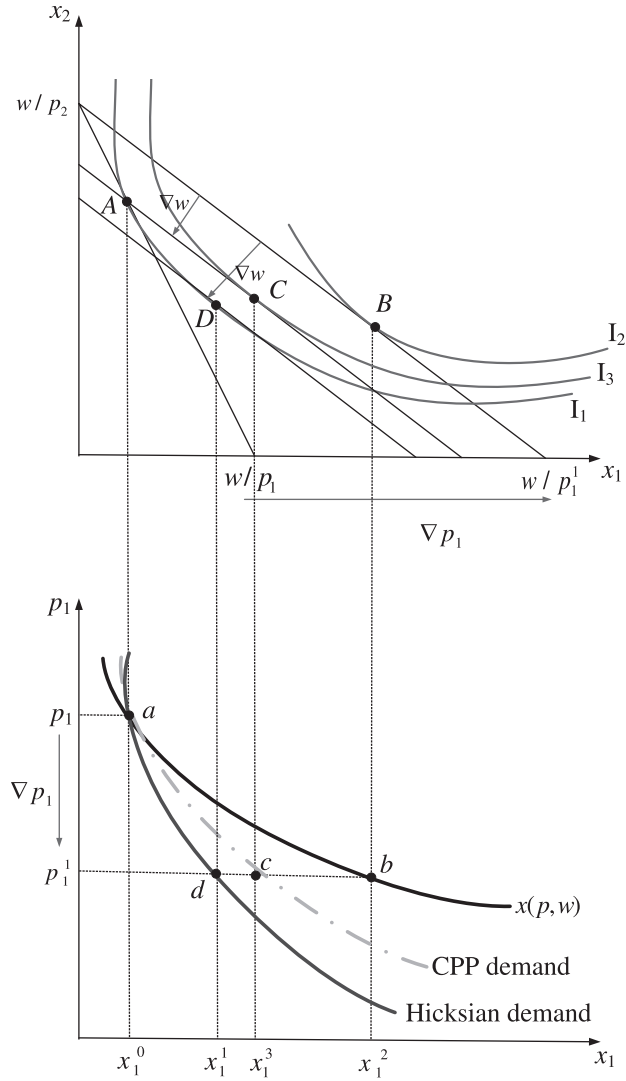


Figure 2.32
Hicksian wealth compensation

a price decrease entails a larger increase in consumption according to the CPP demand, namely from x_1^0 to x_1^3 (as depicted from point a to c), and an even larger increase in the Walrasian demand for the product, namely from x_1^0 to x_1^2 (from point a to b).

In summary, our analysis revealed how different types of demand for x_1 are affected by a given decrease in p_1 . First, a decrease in the price increases the quantity demanded of a good solely due to the price effect (either measured by the Hicksian demand curve or the CPP demand curve). This increase is smaller than the increase in the quantity demanded measured by the Walrasian demand, since this demand measures both price and wealth effects (and this good is not inferior). Second, the wealth compensation (wealth reduction in our example) that maintains the original level of utility of the individual is larger than the wealth compensation that maintains his purchasing power unaltered.¹⁸

Let us now briefly review the Slutsky equation, and the classification of goods as normal, inferior, or Giffen. Figures 2.33 to 2.35 show a decrease in the price of the good on the horizontal axis (food). In all figures, the substitution effect moves in the opposite direction as the price changes (so a reduction in the price of food implies a positive substitution effect). In figure 2.33, the income effect is also positive, thus reinforcing the substitution effect and indicating that this good is normal.¹⁹

In figure 2.34 the substitution effect is still moving in the opposite direction of price (which holds by definition), but the income effect is now negative, partially offsetting the increase in the quantity demanded associated with the substitution effect. Nonetheless, the total effect is still positive. In this case food is an inferior good since the income effect is negative.

Finally, in figure 2.35 the income effect is also negative, but it is now sufficiently large (in absolute value) to completely offset the substitution effect. As a consequence the total effect becomes negative. Graphically, unlike figures 2.33 and 2.34, which generate a negatively sloped Walrasian demand curve, the total effect in figure 2.35 produces a positively sloped Walrasian demand.

¹⁸ Reality check: as an undergraduate student, you probably used the Hicksian wealth compensation when examining income and substitution effects, since, after a price change, you adjusted the consumer's wealth in order for him to reach his initial utility level. However, the Slutsky equation (and the Slutsky matrix) is obtained from applying the Slutsky wealth compensation. An interesting practice is to redo one of the graphical representations of the income and substitution effects of your intermediate microeconomics textbook applying the Slutsky wealth compensation rather than the Hicksian wealth compensation.

¹⁹ Note that BL_d depicts the budget line after applying the Hicksian wealth compensation, which guarantees that the consumer can still reach the same utility level as before the price change.

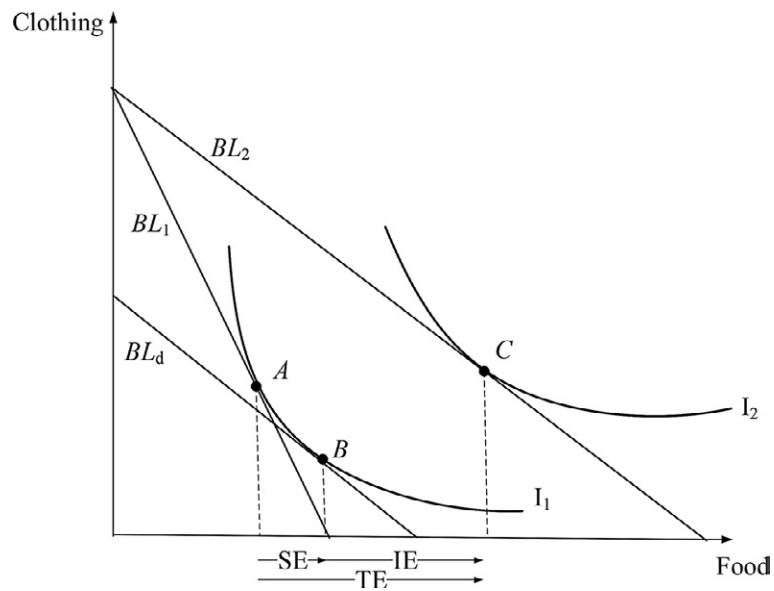


Figure 2.33
Substitution and income effects: Normal goods

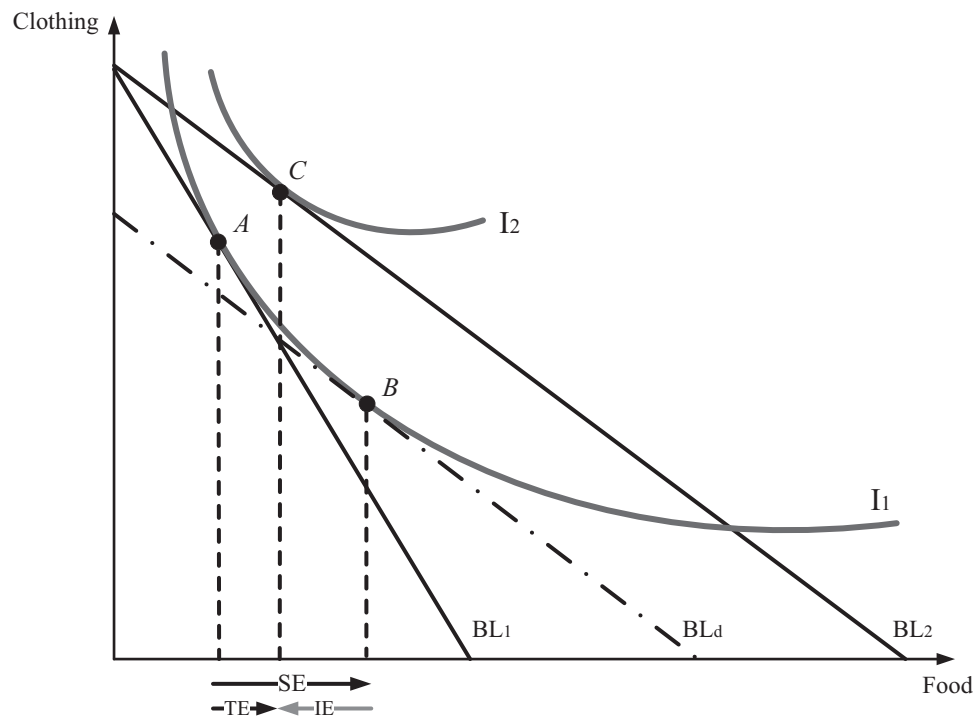


Figure 2.34
Substitution and income effects: Inferior good

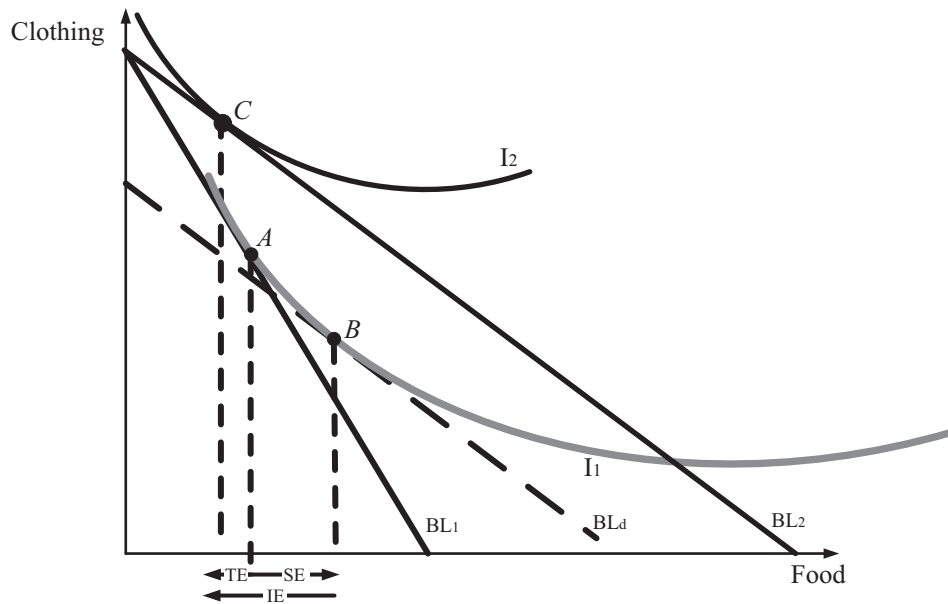


Figure 2.35
Substitution and income effects: Giffen good

Comparing figures 2.34 and 2.35, we can conclude that

Inferior good $\nRightarrow$ Giffen good.

That is, an inferior good does not necessarily have to be a Giffen good. As shown in figure 2.34, for inferior goods the income effect is negative but the total effect is still positive. However, Giffen goods must be inferior, as figure 2.35 shows, since the total effect can only be negative if the income effect is negative (inferior good) and sufficiently large to completely offset the substitution effect.²⁰

2.6 Expenditure Minimization Problem

In the preceding sections we considered an individual decision maker who solves his UMP by choosing a bundle that maximizes his utility level subject to a budget constraint. Alternatively, we could understand the process by which the consumer chooses

²⁰ For more applications of income and substitution effects and the Slutsky equation, see exercises 8–10 and 12–13 at the end of the chapter.

an optimal consumption bundle as a problem in which the consumer *minimizes* his total expenditure on goods and services subject to the constraint that he seeks to reach a given utility level. We refer to this minimization problem as the expenditure minimization problem (EMP). More formally,

$$\min_{x \geq 0} p \cdot x \quad \text{subject to } u(x) \geq u. \quad (\text{EMP})$$

As figure 2.36 indicates, the EMP can be understood as a problem in which the consumer seeks a utility level associated with a particular indifference curve, while spending as little as possible (shifting the budget line, in a parallel fashion, to be as close as possible to the origin). Intuitively, bundles strictly above x^* cannot be a solution to the EMP because, despite reaching the utility level u , they do not minimize total expenditure. Indeed there are other bundles, such as x^* , in which total expenditure is lower and where the consumer still reaches the “target” utility level u . Finally, bundles on the budget line strictly below x^* cannot be the solution to the EMP problem because, despite being cheaper than x^* , the bundles do not satisfy the constraint of reaching utility level u (they reach a lower utility level).

As in the UMP, we can set up the Lagrangian for this minimization problem, and then apply first-order (necessary) conditions with respect to each good and the Lagrange multiplier. Setting up the Lagrangian obtains

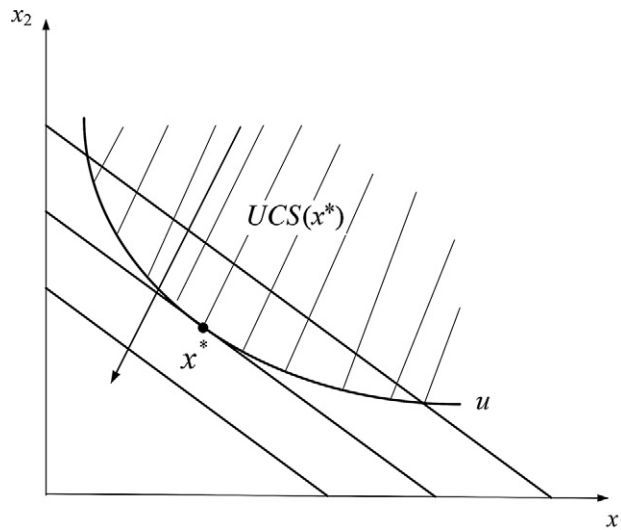


Figure 2.36
Expenditure minimization problem

$$L = p \cdot x + \mu [u - u(x)],$$

and taking first-order (necessary) conditions obtains

$$\frac{\partial L}{\partial x_k} = p_k - \mu \frac{\partial u(x^*)}{\partial x_k} \leq 0$$

for every good k ($=0$ for interior solutions), and

$$\frac{\partial L}{\partial \mu} = u - u(x^*) = 0.$$

Focusing on interior solutions, the first-order conditions above are satisfied with equality. Therefore $p_k = \mu [\partial u(x^*) / \partial x_k]$ for every good k . That is, $1/\mu = [\partial u(x^*) / \partial x_k / p_k]$ for all goods, implying that

$$\frac{\partial u(x^*) / \partial x_k}{p_k} = \frac{\partial u(x^*) / \partial x_l}{p_l}$$

for any two goods l and k , where $l \neq k$. Intuitively, this condition shows that the consumer allocates his consumption across goods until the point where the marginal utility per dollar spent on each good (the “bang for the buck” he obtains) is equal across all goods. Note that the condition above can be alternatively expressed as

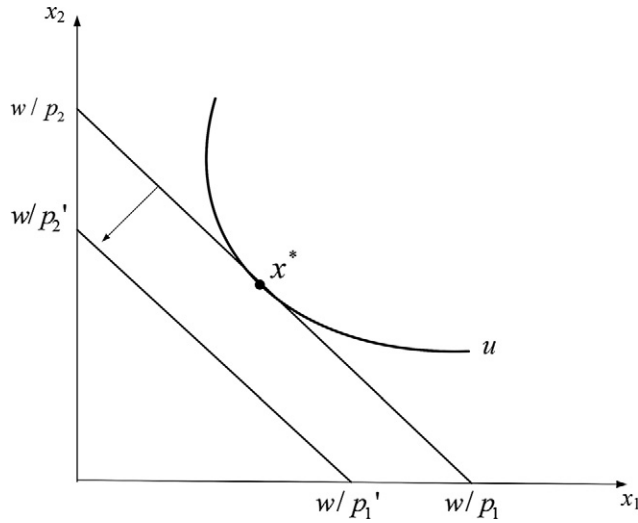
$$\frac{\partial u(x^*) / \partial x_k}{\partial u(x^*) / \partial x_l} = \frac{p_k}{p_l},$$

implying that at the optimum EMP the slope of the indifference curve (the *MRS*) coincides with that of the budget line. Hence, at an interior optimum of the EMP, we obtain the same optimal consumption bundle as when solving the “dual” UMP problem.

2.6.1 Hicksian Demand

The bundles x^* that solve the EMP are denoted as the Hicksian demand, $h(p, u)$, which is a function of the price vector p and the utility level that the individual seeks to reach, u .²¹ Hicksian demand satisfies different properties. In particular, when the utility function is continuous and represents a preference relation satisfying LNS, the Hicksian demand associated to the EMP satisfies:

²¹ As a formality, when the EMP provides more than one solution, we write $x^* \in h(p, u)$, but when the EMP has only one solution, we write $x^* = h(p, u)$.

**Figure 2.37**

Hicksian demand homogeneous of degree zero in prices

1. *Homogeneity of degree zero in prices:* $h(p, u) = h(\alpha p, u)$ for any p, u , and $\alpha > 0$

This property states that, if bundle x is a solution to the EMP when the consumer faces a price vector p , it must also be a solution to the problem when all prices have been scaled by factor α . Intuitively, a common change in all prices does not alter the slope of the consumer's budget line. Figure 2.37 provides a graphical representation. An increase in the prices of both goods, x_1 and x_2 , by the same proportion α produces a downward shift in the budget line. However, the consumer must reach the same utility level u as before the price change in order to satisfy the constraint of the EMP (reach utility level u). Hence he will need to spend more in order to buy bundle x^* at the new prices. As a result he selects the same optimal bundle x^* before and after the price change.²²

2. *No excess utility* For any optimal consumption bundle $x \in h(p, u)$ that solves the EMP, the utility level satisfies $u(x) = u$. Figure 2.38 provides a graphical interpretation of this property. First, consider a bundle $x \in h(p, u)$, for which the consumer obtains a utility level of $u(x) = u^1 > u$. That is, he obtains a utility level higher than what he needs to reach in the EMP. We can then find another bundle x' by scaling

²² As a remark, we define homogeneity of degree zero for all $\alpha > 0$. This implies that a similar argument as the one developed here for any price *increase* in both goods (when $\alpha > 1$) can be extended to a price *decrease* in both goods (when $\alpha < 1$).

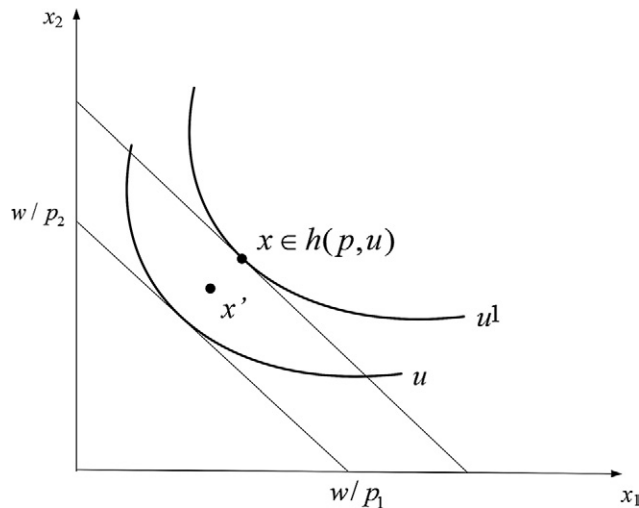


Figure 2.38
No excess utility

down x , so that $x' = \alpha x$ and is very close to x (α close to 1) for which the utility level associated with bundle x' is larger than u , $u(x') > u$. Graphically, this implies that the indifference curve passing through bundle x' is associated with a utility level higher than u but lower than u^1 . Therefore bundle x' exceeds the minimal utility level that the consumer must reach in his EMP, u , and it is cheaper than the original bundle x . As a consequence, for a given utility level u that the consumer seeks in the EMP, bundle $h(p, u)$ does not exceed u , since otherwise he would be able to find a cheaper bundle that exactly reaches utility level u .

3. *Convexity* If the preference relation is convex, then $h(p, u)$ is a convex set. (This property is also related with the following uniqueness property.)
4. *Uniqueness* If the preference relation is strictly convex, then $h(p, u)$ contains a single element. Figure 2.39 describes properties 3 and 4.²³ Uniqueness is a more restrictive concept than convexity, since the former does not necessarily hold when the utility function is linear in both goods (i.e., when the individual regards both goods as perfect substitutes). However, convexity of the optimal bundles in $h(p, u)$ holds both for linear and nonlinear utility functions.

²³ Example 2.4 explains how to find the Hicksian demand for a Cobb–Douglas utility function. It is straightforward to check that the Hicksian demand satisfies homogeneity of degree zero in prices, no excess utility and uniqueness.

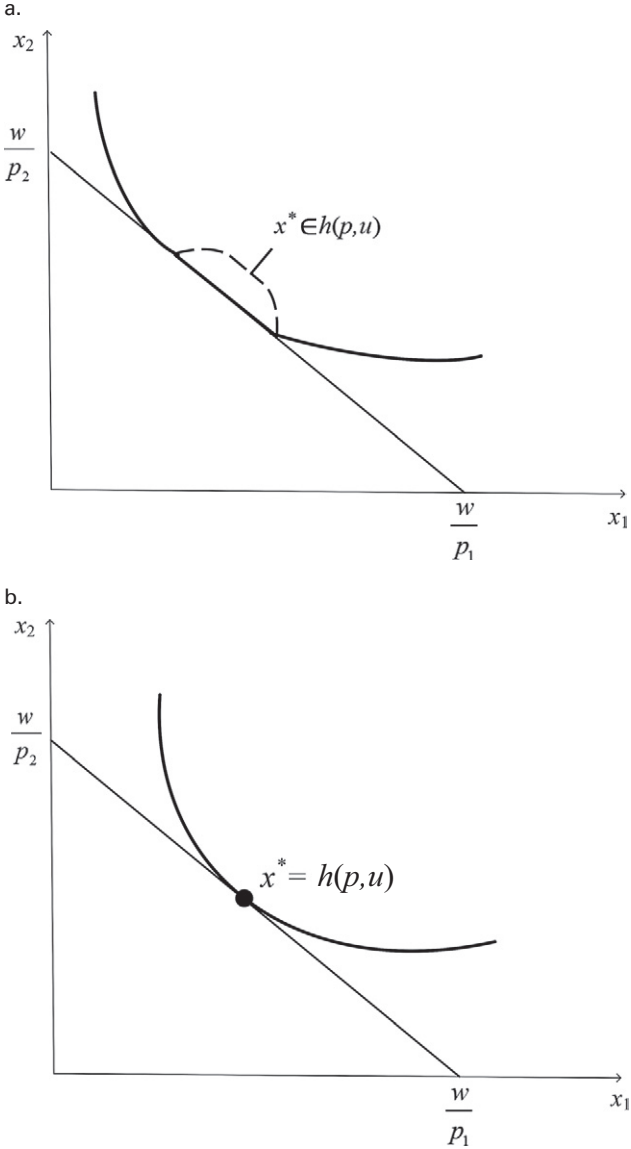


Figure 2.39
Convexity and uniqueness of the Hicksian demand

5. *Compensated law of demand* When the price vector changes from p to p' , the compensated law of demand (CLD) determines that

$$(p' - p) \cdot [h(p', u) - h(p, u)] \leq 0.$$

This property implies that for every good k , Hicksian demand and price move in different directions,

$$(p'_k - p_k) \cdot [h_k(p', u) - h_k(p, u)] \leq 0.$$

Recall that this property holds for the Hicksian demand in which the consumer's wealth is already compensated, so he can still reach his initial utility level ("Hicksian" wealth compensation), as depicted in figure 2.32. However, this property is not necessarily true for the uncompensated Walrasian demand, as described in previous sections of this chapter.

Example 2.4: EMP with a Cobb–Douglas utility function Consider a consumer with Cobb–Douglas utility function $u(x_1, x_2) = x_1^\alpha x_2^\beta$, where $\alpha, \beta > 0$. We can find the Hicksian demands for goods 1 and 2 by solving the following EMP.

Taking first-order conditions with respect to x_1 , x_2 , and μ , yields

$$\min_{x_1, x_2 \geq 0} p_1 x_1 + p_2 x_2 \quad \text{subject to} \quad x_1^\alpha x_2^\beta \geq u.$$

$$p_1 \geq \mu \alpha x_1^{\alpha-1} x_2^\beta,$$

$$p_2 \geq \mu \beta x_1^\alpha x_2^{\beta-1}, \text{ and}$$

$$x_1^\alpha x_2^\beta \geq u$$

where μ denotes the Lagrange multiplier of this minimization problem. In the case of interior solutions, the first-order conditions above become

$$p_1 = \mu \alpha x_1^{\alpha-1} x_2^\beta,$$

$$p_2 = \mu \beta x_1^\alpha x_2^{\beta-1},$$

$$x_1^\alpha x_2^\beta = u.$$

Thus, simultaneously solving for x_1 and x_2 , we find that the Hicksian demands for goods 1 and 2 are

$$h_1(p, u) = \left(\frac{\alpha p_2}{\beta p_1} \right)^{\beta/(\alpha+\beta)} u^{1/(\alpha+\beta)} \quad \text{and} \quad h_2(p, u) = \left(\frac{\beta p_1}{\alpha p_2} \right)^{\alpha/(\alpha+\beta)} u^{1/(\alpha+\beta)}.$$

It is straightforward to show that the Hicksian demands satisfy homogeneity of degree zero in prices, since increasing all prices by the same factor $\lambda > 0$ keeps both demand functions unaffected:

$$h_1(p, u) = \left(\frac{\alpha \lambda p_2}{\beta \lambda p_1} \right)^{\beta/(\alpha+\beta)} u^{1/(\alpha+\beta)} = \left(\frac{\alpha p_2}{\beta p_1} \right)^{\beta/(\alpha+\beta)} u^{1/(\alpha+\beta)},$$

$$h_2(p, u) = \left(\frac{\beta \lambda p_1}{\alpha \lambda p_2} \right)^{\alpha/(\alpha+\beta)} u^{1/(\alpha+\beta)} = \left(\frac{\beta p_1}{\alpha p_2} \right)^{\alpha/(\alpha+\beta)} u^{1/(\alpha+\beta)}.$$

The Hicksian demands are also unique, since for a given price vector, a given utility level u , and a specific value for parameters α and β , we can obtain a precise quantity demanded for goods 1 and 2. Finally, the “no excess utility” property holds, since, in evaluating the Cobb–Douglas utility function at the Hicksian demands, we obtain the exact utility level u that we sought to reach in the EMP. In particular,

$$\begin{aligned} u(h_1(p, u), h_2(p, u)) &= h_1(p, u)^\alpha h_2(p, u)^\beta \\ &= \left[\left(\frac{\alpha p_2}{\beta p_1} \right)^{\beta/(\alpha+\beta)} u^{1/(\alpha+\beta)} \right]^\alpha \left[\left(\frac{\beta p_1}{\alpha p_2} \right)^{\alpha/(\alpha+\beta)} u^{1/(\alpha+\beta)} \right]^\beta \\ &= \left(\frac{\alpha p_2}{\beta p_1} \right)^{\alpha\beta/(\alpha+\beta)} u^{\alpha/(\alpha+\beta)} \left(\frac{\beta p_1}{\alpha p_2} \right)^{\alpha\beta/(\alpha+\beta)} u^{\beta/(\alpha+\beta)} \\ &= \left(\frac{\alpha p_2}{\beta p_1} \frac{\beta p_1}{\alpha p_2} \right) u^{(\alpha+\beta)/(\alpha+\beta)} = u, \end{aligned}$$

as required. ■

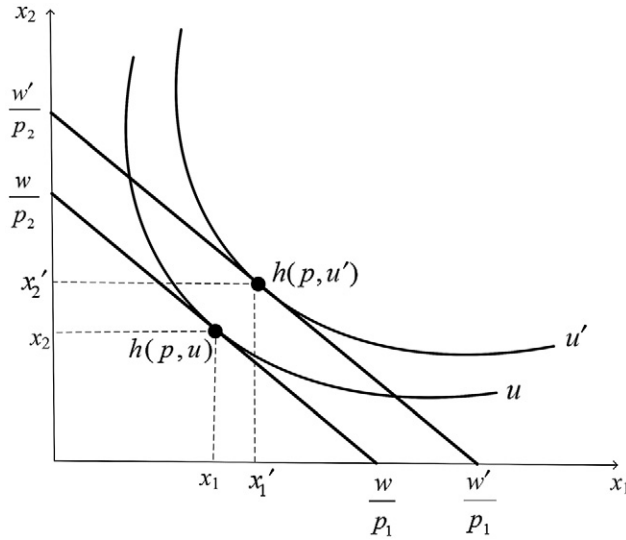
2.6.2 Expenditure Function

Plugging the result of the EMP into the objective function, we obtain the value function of this optimization problem

$$p \cdot h(p, u) = e(p, u),$$

where $e(p, u)$ represents the minimum expenditure that the consumer needs to incur in order to reach utility level u when market prices are p . It is usually denoted as the “expenditure function.” This function also satisfies a set of interesting properties when the utility function is continuous and it represents a preference relation that satisfies LNS:

1. *Homogeneity of degree one in prices* Intuitively, we know that the optimal bundle of the EMP, $h(p, u)$ is not changed when all prices are changing by a common factor λ (because the Hicksian demand satisfies homogeneity of degree zero in prices). Such a change in prices only makes it more or less expensive to buy the same bundle. That is, $e(\lambda p, u) = \lambda e(p, u)$. For instance, a 50 percent increase in all prices makes the optimal consumption bundle 50 percent more expensive.

**Figure 2.40**

Expenditure function $e(p, u)$ strictly increasing in u

2. *Strictly increasing in u* For a given price vector, reaching a higher utility level u requires an increase in one's expenditure, as figure 2.40 indicates.

When the consumer solves an EMP that reaches utility level u , his Hicksian demand is represented by bundle $h(p, u) = (x_1, x_2)$, with associated expenditure function $e(p, u) = p_1x_1 + p_2x_2$. However, when this consumer must reach a higher utility level u' , he solves a new EMP where he chooses Hicksian demand $h(p, u') = (x_1', x_2')$, where $x_k' > x_k$ for at least one good $k = \{1, 2\}$, with associated expenditure function $e(p, u') = p_1x_1' + p_2x_2'$. Comparing the expenditure in both settings obtains

$$p_1x_1 + p_2x_2 < p_1x_1' + p_2x_2'.$$

The price vector is unaffected, but the consumer is purchasing larger amounts of good 1, good 2, or both, in order to reach a higher utility level u' . Hence the expenditure function satisfies $e(p, u') > e(p, u)$, implying that $e(p, u)$ is strictly increasing in u .

3. *Weakly increasing in prices, for any good k* Higher prices mean that the consumer will need to incur a larger expenditure in order to reach the same utility level.

Proof Consider two price vectors p and p' , where

$$p = (p_1, p_2, \dots, p_k, \dots, p_L), \text{ and } p' = (p_1, p_2, \dots, p'_k, \dots, p_L),$$

where $p'_k \geq p_k$. That is, we only increase the price of good k , and leave the prices of all other goods unaffected. Let $x = h(p, u)$ denote the Hicksian demand arising from the EMP under price vector p , and similarly let $x' = h(p', u)$ be the Hicksian demand under price vector p' . Then the associated expenditure functions are

$$p \cdot x = e(p, u) \quad \text{and} \quad p' \cdot x' = e(p', u).$$

Hence

$$e(p', u) = p' \cdot x' \geq p \cdot x' \geq p \cdot x = e(p, u),$$

where the first inequality is explained by the fact that $p' \geq p$ (since $p'_k \geq p_k$ for good k and $p'_l = p_l$ for all the other goods $l \neq k$). The second inequality is rationalized by the fact that, at a given price vector p , the bundle that solves the EMP for price vector p is $x = h(p, u)$. Hence any other bundle $x' \neq x$ is not expenditure minimizing and must therefore be more costly than x . Thus $e(p', u) \geq e(p, u)$, implying that the expenditure function is weakly increasing in the price of any good k .²⁴ ■

4. *Concave in prices* We provide a graphical representation of this property in figure 2.41, which we next use in the proof of concavity.

Proof Let us start from bundle x' , where $x' \in h(p', u)$. By definition, we know that the total expenditure at price vector p' is lower (or equal) when buying bundle x' than when buying any other bundle $x \neq x'$. For instance, this is true for bundle $\bar{x}$, where $p'x' \leq p'\bar{x}$. Similarly, at prices p'' , the total expenditure $e(p'', u)$ when buying bundle x'' , where $x'' \in h(p'', u)$, is lower (or equal) than from buying any other bundle $x \neq x''$. For instance, bundle $\bar{x}$ entails an expenditure $p''\bar{x}$ where $p''x'' \leq p''\bar{x}$. Using the results found above for prices p' and p'' , we find that

$$\text{if } p'x' \leq p'\bar{x} \quad \text{and} \quad p''x'' \leq p''\bar{x}.$$

Then their convex combination must satisfy

$$\alpha p'x' + (1 - \alpha)p''x'' \leq \alpha p'\bar{x} + (1 - \alpha)p''\bar{x}.$$

Rearranging, and defining $\bar{p} \equiv \alpha p' + (1 - \alpha)p''$, we obtain

$$\alpha p'x' + (1 - \alpha)p''x'' \leq [\alpha p' + (1 - \alpha)p'']\bar{x} = \bar{p}\bar{x},$$

which ultimately entails

²⁴ Recall that homogeneity tests how much the expenditure function increases when the price of all goods increases, whereas this property only considered the increase in the price of good k .

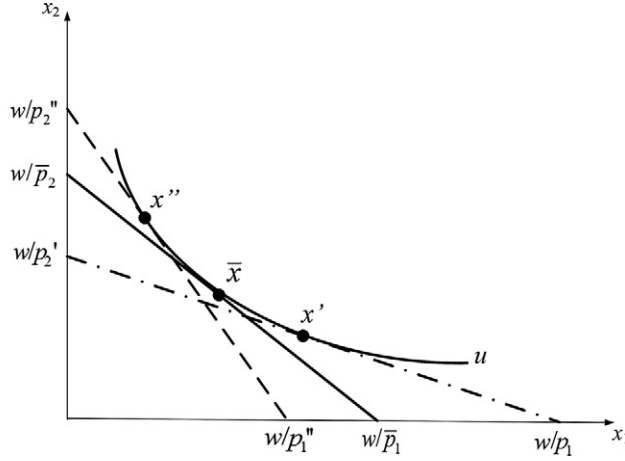


Figure 2.41
Expenditure concave in prices

$$\alpha e(p', u) + (1 - \alpha) e(p'', u) \leq e(\bar{p}, u).$$

Therefore the expenditure function is concave in prices. ■

Intuitively, the convex combination of price vector p' and p'' yields an intermediate price vector $\bar{p}$, with an associated expenditure function $e(\bar{p}, u)$ that is more costly than either the optimal bundle under prices p' , $x' \in h(p', u)$ with expenditure $e(p', u)$, or the optimal bundle under prices p'' , $x'' \in h(p'', u)$ with expenditure $e(p'', u)$.

Example 2.5: Expenditure function of a Cobb–Douglas utility function For the Cobb–Douglas utility function $u(x_1, x_2) = x_1^\alpha x_2^\beta$ considered above, we know that the Hicksian demands for goods 1 and 2 are

$$h_1(p, u) = \left(\frac{\alpha p_2}{\beta p_1} \right)^{\beta/(\alpha+\beta)} u^{1/(\alpha+\beta)}, \quad h_2(p, u) = \left(\frac{\beta p_1}{\alpha p_2} \right)^{\alpha/(\alpha+\beta)} u^{1/(\alpha+\beta)}.$$

Therefore the expenditure function is

$$\begin{aligned} e(p, u) &= p_1 \cdot h_1(p, u) + p_2 \cdot h_2(p, u) \\ &= p_1 \cdot \left[\left(\frac{\alpha p_2}{\beta p_1} \right)^{\beta/(\alpha+\beta)} u^{1/(\alpha+\beta)} \right] + p_2 \cdot \left[\left(\frac{\beta p_1}{\alpha p_2} \right)^{\alpha/(\alpha+\beta)} u^{1/(\alpha+\beta)} \right], \end{aligned}$$

and it can be simplified as

$$e(p, u) = p_1^{\alpha/(\alpha+\beta)} p_2^{\beta/(\alpha+\beta)} u^{1/(\alpha+\beta)} \left[\left(\frac{\alpha}{\beta} \right)^{\beta/(\alpha+\beta)} + \left(\frac{\beta}{\alpha} \right)^{\alpha/(\alpha+\beta)} \right],$$

which is homogeneous of degree one in prices, strictly increasing in u . In addition, $e(p, u)$ is strictly increasing in the prices of every good k , p_k , and at a decreasing rate, thus satisfying concavity in prices. ■

2.7 Relationships between the Expenditure Function and Hicksian Demand

Consider a continuous utility function representing a preference relation that satisfies LNS and that is strictly convex. Moreover, for all p and u , we have

$$\frac{\partial e(p, u)}{\partial p_k} = h_k(p, u) \quad \text{for every good } k.$$

This identity (often referred to as “Shepard’s lemma”) tells us that if we want to find the Hicksian demand for good k , $h_k(p, u)$, and we have information about the expenditure function, we just need to differentiate $e(p, u)$ with respect to p_k .²⁵ Note that this result is convenient from a practical point of view. In particular, if one does not know the actual expression of the expenditure function, or if its expression is relatively intractable, we can still measure the reaction of a consumer’s minimal expenditure to changes in prices, $\partial e(p, u) / \partial p_k$, by using his Hicksian demand for that good, $h_k(p, u)$.

The relationship between the Hicksian demand and the expenditure function can be further developed by taking first-order conditions again. That is, $\partial^2 e(p, u) / \partial p_k^2 = \partial h_k(p, u) / \partial p_k$ or in matrix notation $D_p^2 e(p, u) = D_p h(p, u)$. And since $D_p h(p, u)$ provides the Slutsky matrix $S(p, w)$, an immediate implication of this result is that

$$S(p, w) = D_p^2 e(p, u).$$

This is a useful identity, since the Slutsky matrix can be obtained from the observable Walrasian demand. From the Slutsky matrix we can therefore understand how a change in prices affects the Hicksian demand for a particular good. There are three other important properties of $D_p h(p, u)$:

1. $D_p h(p, u)$ is *negative semidefinite*. Therefore the Slutsky matrix is also negative semidefinite, a property we discussed in previous sections.

²⁵ We can prove this result using three different approaches: the duality theorem, first-order conditions, or the envelope theorem. All three proofs are presented in the appendix B at the end of the chapter.

2. $D_p h(p, u)$ is a symmetric matrix. Therefore the Slutsky matrix is symmetric.²⁶
3. $D_p h(p, u)p = 0$, which implies that $S(p, w)p = 0$. From an economic point of view, this property implies that all goods cannot be net substitutes in consumption nor can they all be net complements. In particular, we refer to two goods as net substitutes (net complements) when the derivative of the Hicksian demand of good l , with respect to the price of good k , is positive (negative, respectively). If all goods were net substitutes (all were net complements), all the terms away from the main diagonal in the $D_p h(p, u)$ matrix would be positive (negative, respectively). In this case the multiplication of this vector of derivatives times the positive price vector $p \gg 0$ would yield a nonzero result.

2.8 Relationship between the Walrasian and Hicksian Demand

As described above, when income effects are positive (normal goods), the Walrasian demand is above the Hicksian demand as figure 2.42 illustrates.²⁷

Before the price decrease the representative consumer selects bundle x^* , and after the price decrease, the consumer selects bundle x_2 . Figure 2.42 illustrates the Hicksian wealth compensation by which the individual's wealth level is reduced so that he can reach his initial utility (reaches his initial indifference curve I_1) at the new price ratio. That is, the budget line after the price change is shifted inward until the point where it becomes tangent to the initial indifference curve. Given this reduced wealth level, the consumer selects bundle x_1 . Therefore the increase in the consumption of good 1 reflected by the movement from x^* to x^1 denotes the *substitution effect*. Only the price ratio has changed, and the consumer's wealth level allows him to still afford bundles that provide the same utility as initial bundle x^* . However, the increase in consumption of good 1 from x^1 to x^2 reflects the *income effect*, since, for a given price ratio, a larger purchasing power allows the consumer to increase his consumption of good 1.

In the bottom panel of figure 2.42 we summarize our previous discussion. In particular, the increase in the consumption of good 1 from x_1^* to x_1^2 (capturing both the substitution and income effects) is represented by the Walrasian demand (i.e., movement from a to b). In contrast, when we measure the substitution effect associated with

²⁶ From our previous discussion, we could not conclude that the Slutsky matrix was symmetric. In our analysis of the Hicksian demand, we have assumed that the utility function is continuous and preferences are strictly convex, which provides us now with the result that $S(p, w)$ is symmetric.

²⁷ As in previous sections, figure 2.42 represents a decrease in the price of good one. The analysis of a price increase is analogous.

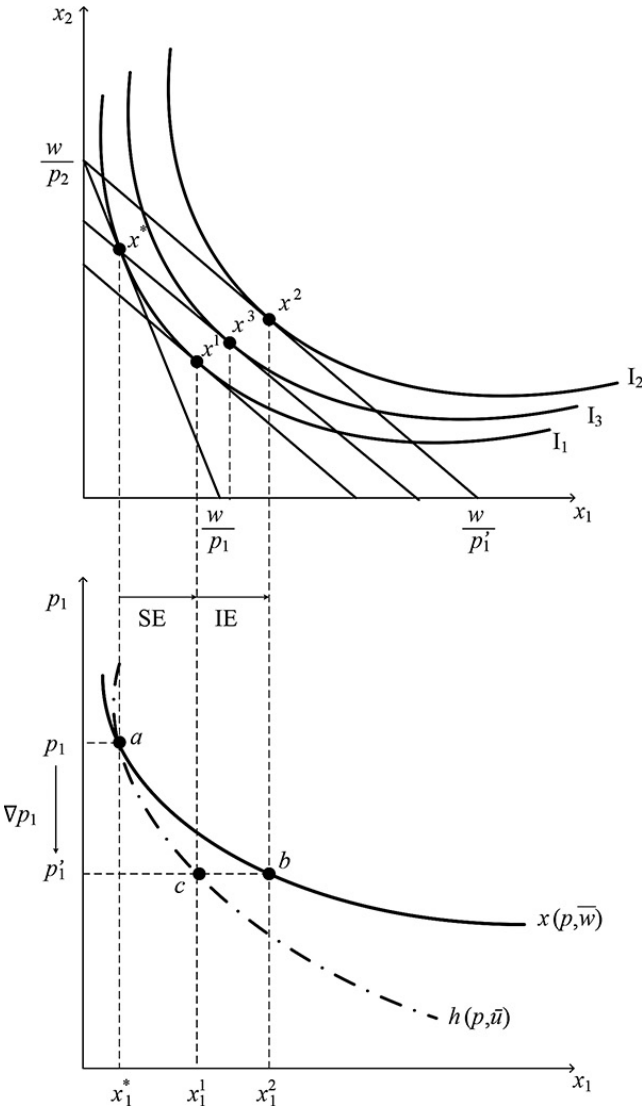


Figure 2.42
Walrasian and Hicksian demand: Normal good

the price change, we use the Hicksian demand. Indeed, for the same decrease in p_1 , the Hicksian demand reflects a smaller increase in the consumption of x_1 than the Walrasian demand, given that it only captures the substitution effect (i.e., movement from a to c along the Hicksian demand). We can therefore conclude that, for a given price decrease, the Walrasian demand is *flatter* than the Hicksian demand when the good that we analyze is normal in consumption.²⁸

In contrast, when income effects are negative (so that the consumer regards a good as inferior), the Walrasian demand is steeper than the Hicksian demand. Figure 2.43 illustrates a decrease in p_1 , where x_1 is regarded as inferior.

First, note that the substitution effect associated with the price decrease is still positive, from bundle x^* to x^1 , as in figure 2.42. The income effect in this case, however, is negative, reducing the consumption of good 1 from x^1 to x^2 . Therefore the substitution effect is ameliorated by the negative income effect, leading to a relatively small total effect. If we map these bundles into the bottom panel of figure 2.43, we see that the total effect represented on the top panel of figure 2.43 is relatively minor, which implies that the increase in the quantity demanded captured by the Walrasian demand is also relatively small. However, the substitution effect (represented in the Hicksian demand) is substantial, leading to a relatively flat Hicksian demand curve. Therefore, when a good is inferior, we can expect that a given decrease in its price produces a larger increase in the quantity demanded under Hicksian demand than under Walrasian demand.

We can now formally relate the Hicksian and Walrasian demands using the Slutsky equation. For consistency, we consider a continuous utility function, representing a strictly convex preference relation that satisfies LNS, and a consumer facing prices and wealth $(\bar{p}, \bar{w})$ who attains a utility level $\bar{u}$. Then, his wealth satisfies $\bar{w} = e(\bar{p}, \bar{u})$. In addition we know that for any (p, u) pair, the Hicksian demand for good l satisfies, $h_l(p, u) = x_l(p, e(p, u))$. Differentiating this expression with respect to p_k , and evaluating it at $(\bar{p}, \bar{u})$, we obtain

$$\frac{\partial h_l(\bar{p}, \bar{u})}{\partial p_k} = \frac{\partial x_l(\bar{p}, e(\bar{p}, \bar{u}))}{\partial p_k} + \frac{\partial x_l(\bar{p}, e(\bar{p}, \bar{u}))}{\partial w} \frac{\partial e(\bar{p}, \bar{w})}{\partial p_k}.$$

Using Shephard's lemma, $\partial e(\bar{p}, \bar{u})/\partial p_k = h_k(\bar{p}, \bar{u})$, we can write

²⁸ For completeness, figure 2.42 also depicts the Slutsky wealth compensation that reduces the individual's wealth level so that he can afford his initial bundle. After the wealth compensation, the individual selects bundle x^3 . If we wanted to map bundle x^3 into the bottom panel of figure 2.42 we would obtain the "constant purchasing power" demand curve discussed in previous sections of this chapter.

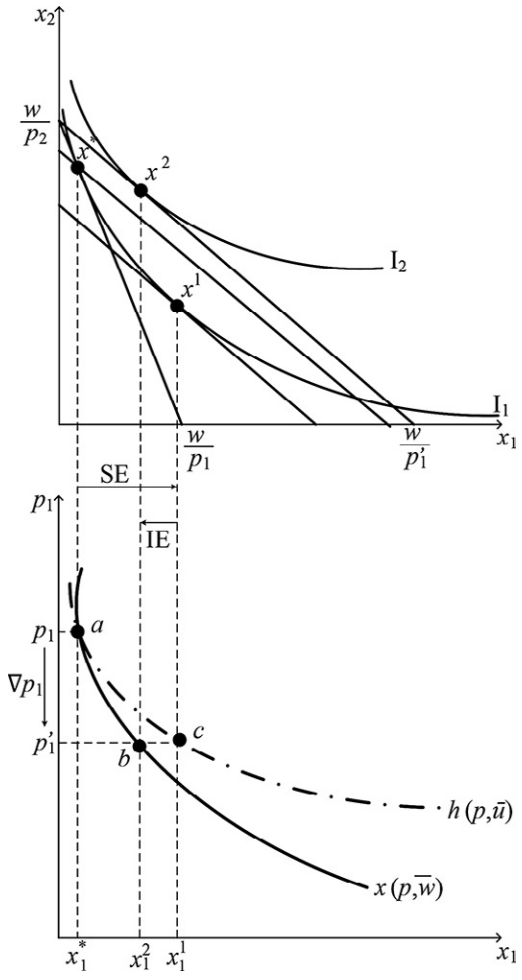


Figure 2.43
Walrasian and Hicksian demand: Inferior good

$$\frac{\partial h_l(\bar{p}, \bar{u})}{\partial p_k} = \frac{\partial x_l(\bar{p}, e(\bar{p}, \bar{u}))}{\partial p_k} + \frac{\partial x_l(\bar{p}, e(\bar{p}, \bar{u}))}{\partial w} h_k(p, u).$$

Finally, since $\bar{w} = e(\bar{p}, \bar{u})$ and $h_k(\bar{p}, \bar{u}) = x_k(\bar{p}, e(\bar{p}, \bar{u})) = x_k(\bar{p}, \bar{w})$, the above expression becomes

$$\frac{\partial h_l(\bar{p}, \bar{u})}{\partial p_k} = \frac{\partial x_l(\bar{p}, \bar{w})}{\partial p_k} + \frac{\partial x_l(\bar{p}, \bar{w})}{\partial w} x_k(\bar{p}, \bar{w}).$$

This expression in fact coincides with term $s_{lk}(p, w)$ as we saw in our discussion of the Slutsky equation. Therefore the matrix of partial derivatives of the Hicksian demand, $D_p h(p, u)$, coincides with the Slutsky matrix.²⁹ Or more compactly, $SE = TE + IE$ or, after rearranging,

$$TE = SE - IE.$$

Hence, given that $\partial h_l(\bar{p}, \bar{u})/\partial p_k < 0$ (i.e., negative SE), the Walrasian demand is more sensitive to such a price change, $|\partial x_l(\bar{p}, \bar{w})/\partial p_k| > |\partial h_l(\bar{p}, \bar{u})/\partial p_k|$, only if the income effect emphasizes the substitution effect, that is, $[\partial x_l(\bar{p}, \bar{w})/\partial w] x_k(\bar{p}, \bar{w}) > 0$.

Further reading For more extensions on the basic demand model, including empirical applications, see chapters 3 and 4 in Deaton and Muellbauer's (1980) textbook.

2.9 Relationship between the Walrasian Demand and the Indirect Utility Function

Consider a continuous utility function representing a strictly convex preference relation that satisfies LNS. Additionally suppose that the indirect utility function associated with the UMP problem is differentiable at any pair of prices and wealth. We then obtain Roy's identity³⁰:

$$-\frac{\partial v(p, w)/\partial p_k}{\partial v(p, w)/\partial w} = x_k(p, w) \quad \text{for every good } k.$$

²⁹ Recall that both of these matrices have $N \times N$ dimension, since they reflect both own- and cross-price effects among all goods.

³⁰ For a proof of Roy's identity, see exercise 28 at the end of this chapter.

This is an interesting result because in certain cases it will be easier to compute the derivatives of the indirect utility function than solving the UMP with the system of first-order conditions, especially when dealing with many goods.

2.10 Summary of Relationships

Let us next briefly summarize the previous relationships between elements of the UMP and the EMP.

1. The Walrasian demand, $x(p, w)$, is the solution of the UMP. The Hicksian demand, $h(p, u)$, is the solution of the EMP. Their associated value functions are, respectively, the indirect utility function, $v(p, w)$, and the expenditure function $e(p, u)$.
2. The value functions of both optimization problems exhibit the following relationships:
 - $e(p, v(p, w)) = w$ In words, the minimal expenditure to reach a utility level equal to the maximal utility that the individual reaches at his UMP, $u = v(p, w)$, must be w .
 - $v(p, e(p, u)) = u$ The indirect utility that the consumer reaches when he is endowed with a wealth level equal to the minimal expenditure he optimally uses in the EMP, $w = e(p, u)$, is exactly u .

The solutions of the UMP and the EMP, and their associated value functions, are illustrated in figure 2.44 with arrow 1 for the UMP (left-hand side of the figure) and EMP (right-hand side), and arrow 2 showing the relationship between the value function of both optimization problems.

3. Let us now connect the argmax of the UMP (the Walrasian demand) with the argmin of the EMP (the Hicksian demand):
 - $x(p, e(p, u)) = h(p, u)$ The (uncompensated) Walrasian demand of a consumer endowed with an adjusted wealth level w (equal to the expenditure he optimally uses in the EMP), $w = e(p, u)$, coincides with his Hicksian demand, $h(p, u)$.
 - $h(p, v(p, w)) = x(p, w)$ The (compensated) Hicksian demand of a consumer reaching the maximum utility of the UMP, $u = v(p, w)$, coincides with his Walrasian demand.

Figure 2.45 superimposes the relationship between the Walrasian and Hicksian demands at the top of figure 2.44. Specifically, both demands are related via the value function of the other optimization problem. For instance, arrow 3a indicates that the Walrasian demand $x(p, w)$ is related to the Hicksian demand $h(p, u)$ via the EMP's value

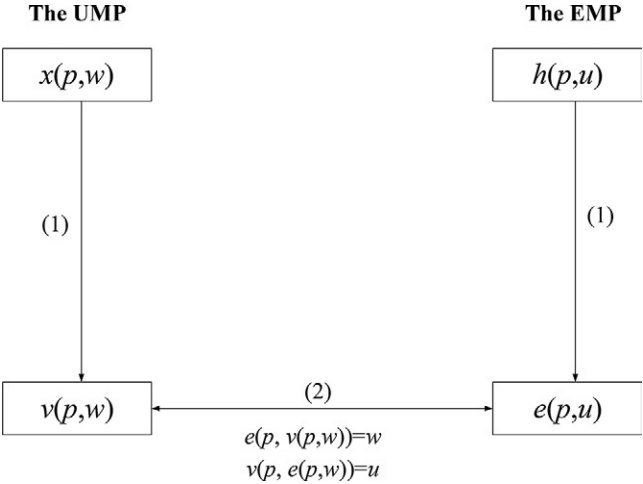


Figure 2.44
UMP and the EMP—I

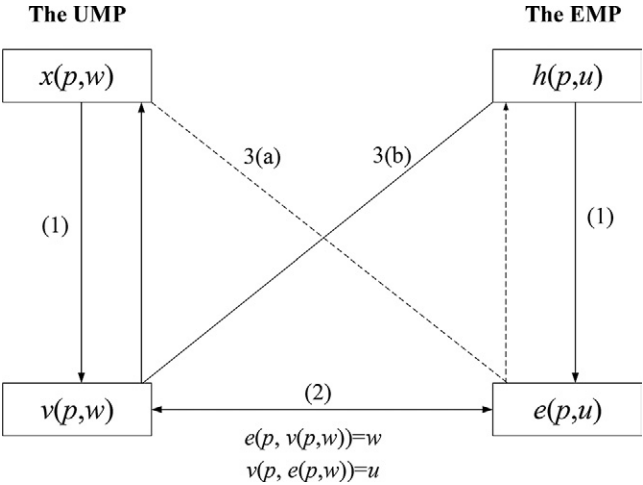


Figure 2.45
UMP and the EMP—II

function, the expenditure function, $e(p, u)$, as indicated in the first point above. A similar argument applies to the Hicksian demand, as depicted in arrow 3b.

Additionally the Walrasian and Hicksian demands can be directly related by the Slutsky equation (recall, however, that this equation relates their derivatives, not their absolute values). Figure 2.46 introduces the Slutsky equation by connecting the derivatives of the Walrasian and Hicksian demands at the top of figures 2.44 and 2.45; see upper part of figure 2.46, at arrow 4a.

Finally, figure 2.47 introduces Roy's identity in connecting the indirect utility function and the Walrasian demand (on the left-hand side of the figure), with arrow 4b, since the Roy's identity states that

$$-\frac{\partial v(p, w)/\partial p_k}{\partial v(p, w)/\partial w} = x_k(p, w).$$

Figure 2.47 also includes Shephard's lemma, by which the derivative of the expenditure function with respect to prices leads to the Hicksian demand, that is, $\partial e(p, u)/\partial p = h(p, u)$, as indicated by arrow 4c on the right-hand side of the figure.

Example 2.6: Duality with a Cobb–Douglas utility function Consider a consumer with Cobb–Douglas utility function $u(x_1, x_2) = x_1^\alpha x_2^\beta$, where $\alpha, \beta > 0$. By duality, we know that the following identities must be true:

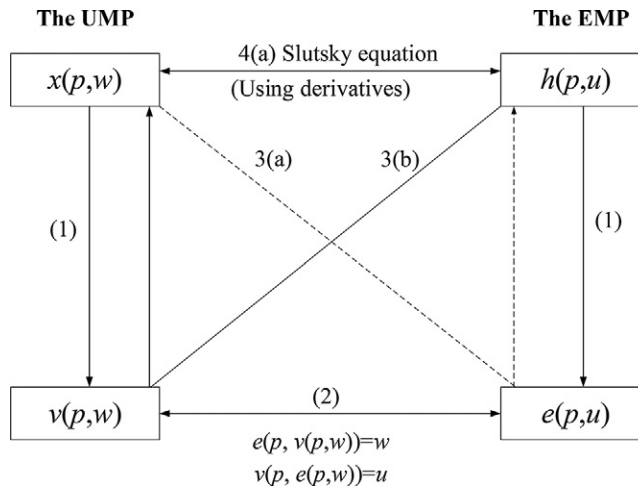


Figure 2.46
UMP and the EMP—III

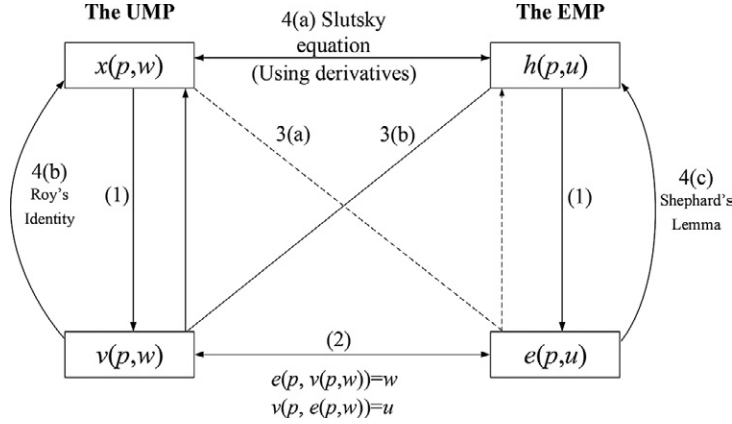


Figure 2.47
UMP and the EMP—IV

$$e(p, v(p, w)) = w \text{ and } v(p, e(p, u)) = u.$$

$$x(p, e(p, u)) = h(p, u) \text{ and } h(p, v(p, w)) = x(p, w).$$

To show that these identities hold for the Cobb–Douglas utility function presented above, we first recall that the Walrasian demands for goods 1 and 2 are

$$x_1(p, w) = \frac{\alpha w}{(\alpha + \beta)p_1} \text{ and } x_2(p, w) = \frac{\beta w}{(\alpha + \beta)p_2},$$

which yields an indirect utility function

$$v(p, w) = \left(\frac{\alpha w}{(\alpha + \beta)p_1} \right)^\alpha \left(\frac{\beta w}{(\alpha + \beta)p_2} \right)^\beta = \left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{\beta}{p_2} \right)^\beta \left(\frac{w}{\alpha + \beta} \right)^{\alpha + \beta}.$$

The Hicksian demands that emerge from the EMP are

$$h_1(p, u) = \left(\frac{\alpha p_2}{\beta p_1} \right)^{\beta/(\alpha + \beta)} u^{1/(\alpha + \beta)}, \quad h_2(p, u) = \left(\frac{\beta p_1}{\alpha p_2} \right)^{\alpha/(\alpha + \beta)} u^{1/(\alpha + \beta)},$$

implying that the expenditure function is

$$\begin{aligned} e(p, u) &= p_1 \cdot \left[\left(\frac{\alpha p_2}{\beta p_1} \right)^{\beta/(\alpha + \beta)} u^{1/(\alpha + \beta)} \right] + p_2 \left[\left(\frac{\beta p_1}{\alpha p_2} \right)^{\alpha/(\alpha + \beta)} u^{1/(\alpha + \beta)} \right] \\ &= p_1^{\alpha/(\alpha + \beta)} p_2^{\beta/(\alpha + \beta)} u^{1/(\alpha + \beta)} \left[\left(\frac{\alpha}{\beta} \right)^{\beta/(\alpha + \beta)} + \left(\frac{\beta}{\alpha} \right)^{\alpha/(\alpha + \beta)} \right]. \end{aligned}$$

Let us start examining the identities in step 2 of figure 2.47, where $e(p, v(p, w)) = w$ and $v(p, e(p, u)) = u$. First, plugging the indirect utility function $v(p, w)$ from the UMP into the expenditure function found in the EMP, $e(p, u)$, yields an expression that can be reduced to w since

$$\begin{aligned} e(p, v(p, w)) &= p_1^{\alpha/(\alpha+\beta)} p_2^{\beta/(\alpha+\beta)} \left(\underbrace{\left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{\beta}{p_2} \right)^\beta \left(\frac{w}{\alpha+\beta} \right)^{\alpha+\beta}}_u \right)^{1/(\alpha+\beta)} \left[\left(\frac{\alpha}{\beta} \right)^{\beta/(\alpha+\beta)} + \left(\frac{\beta}{\alpha} \right)^{\alpha/(\alpha+\beta)} \right] \\ &= \frac{\alpha w}{\alpha+\beta} + \frac{\beta w}{\alpha+\beta} = w. \end{aligned}$$

Likewise, by plugging the expenditure function $e(p, u) = w$ into the indirect utility function $v(p, w)$, we obtain an expression that can be simplified to u , as follows:

$$\begin{aligned} v(p, e(p, u)) &= \left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{\beta}{p_2} \right)^\beta \left\{ \frac{p_1^{\alpha/(\alpha+\beta)} p_2^{\beta/(\alpha+\beta)} u^{1/(\alpha+\beta)} \left[(\alpha/\beta)^{\beta/(\alpha+\beta)} + (\beta/\alpha)^{\alpha/(\alpha+\beta)} \right]}{\alpha+\beta} \right\}^{\alpha+\beta} \\ &= \alpha^\alpha \beta^\beta u \cdot \frac{(\alpha+\beta)^{\alpha+\beta}}{(\alpha+\beta)^{\alpha+\beta} \alpha^\alpha \beta^\beta} = u. \end{aligned}$$

Let us now separately check the two identities in step 3 of figure 2.47. First, from step 3a, we need to have $x(p, e(p, u)) = h(p, u)$. Therefore, by inserting the expenditure function $e(p, u) = w$ into the Walrasian demand of good 1, $x_1(p, w)$, we obtain

$$\begin{aligned} x_1(p, e(p, u)) &= \frac{\alpha p_1^{\alpha/(\alpha+\beta)} p_2^{\beta/(\alpha+\beta)} u^{1/(\alpha+\beta)} \left[(\alpha/\beta)^{\beta/(\alpha+\beta)} + (\beta/\alpha)^{\alpha/(\alpha+\beta)} \right]}{(\alpha+\beta)p_1} \\ &= \left(\frac{\alpha p_2}{\beta p_1} \right)^{\beta/(\alpha+\beta)} u^{1/(\alpha+\beta)} = h_1(p, u). \end{aligned}$$

Similarly plugging $e(p, u) = w$ into the Walrasian demand of good 2 yields

$$\begin{aligned} x_2(p, e(p, u)) &= \frac{\beta p_1^{\alpha/(\alpha+\beta)} p_2^{\beta/(\alpha+\beta)} u^{1/(\alpha+\beta)} \left[(\alpha/\beta)^{\beta/(\alpha+\beta)} + (\beta/\alpha)^{\alpha/(\alpha+\beta)} \right]}{(\alpha+\beta)p_2} \\ &= \left(\frac{\beta p_1}{\alpha p_2} \right)^{\alpha/(\alpha+\beta)} u^{1/(\alpha+\beta)} = h_2(p, u), \end{aligned}$$

which coincides with the Hicksian demand $h(p, u)$. From step 3b, we need $h(p, v(p, w)) = x(p, w)$. Plugging the indirect utility function $v(p, w) = u$ into the Hicksian demand for good 1 yields

$$\begin{aligned}
 h_1(p, v(p, w)) &= \left(\frac{\alpha p_2}{\beta p_1} \right)^{\beta/(\alpha+\beta)} \left(\left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{\beta}{p_2} \right)^\beta \left(\frac{w}{\alpha+\beta} \right)^{\alpha+\beta} \right)^{1/(\alpha+\beta)} \\
 &= \frac{\alpha w}{(\alpha+\beta)p_1} = x_1(p, w).
 \end{aligned}$$

And operating similarly with the Hicksian demand of good 2, we obtain

$$\begin{aligned}
 h_2(p, v(p, w)) &= \left(\frac{\beta p_1}{\alpha p_2} \right)^{\alpha/(\alpha+\beta)} \left(\left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{\beta}{p_2} \right)^\beta \left(\frac{w}{\alpha+\beta} \right)^{\alpha+\beta} \right)^{1/(\alpha+\beta)} \\
 &= \frac{\beta w}{(\alpha+\beta)p_2} = x_2(p, w),
 \end{aligned}$$

which coincides with the Walrasian demand, $x(p, w)$. ■

Appendix A: Duality in Consumption

In this chapter we discussed both the utility maximization problem (UMP), the so-called primal problem describing the consumer's choice of optimal consumption bundles, and its dual: the expenditure minimization problem (EMP). However, when can we guarantee that the solution x^* to both problems coincide?

Rubinstein (2012) provides an intuitive (and humorous) approach to this question. Consider a function $M(t)$ that describes the distance that a turtle travels in time t , as figure A2.1 illustrates.

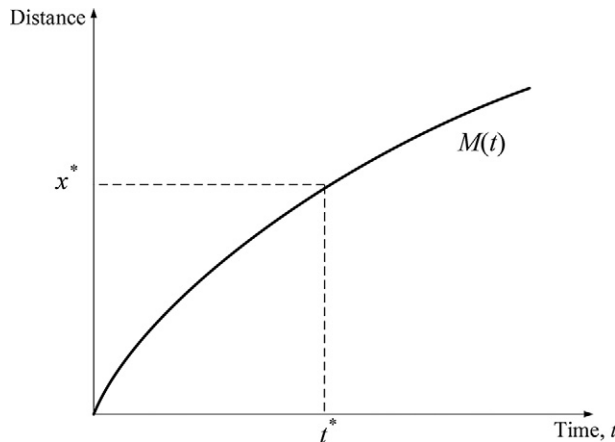


Figure A2.1
Primal and dual problems with a monotonic function